A PROOF OF PILZ'S EXTENDED 1–2–3 CONJECTURE

ABSTRACT. Pilz's Extended 1-2-3 Conjecture asserts that for every finite nonempty set A of positive integers and every $n \geq 1$, at least n integers have an odd number of representations aj with $a \in A$ and $1 \leq j \leq n$; equivalently, $|A \triangle 2A \triangle \cdots \triangle nA| \geq n$. The question goes back to Pilz's work on the minimal distance of near-ring codes. It was previously known for $n \leq 8$, for $|A| \leq 3$, for $A = \{1, \ldots, k\}$, and for all $n \geq 2(3^{80} - 321)$. We prove it for every n.

Encoding each prime p as a variable x_p over $\mathbb{F}_2$ and each integer a as the monomial $X(a) = \prod_p x_p^{v_p(a)}$, the conjecture says that no nonzero multiple of $F_n = \sum_{a \leq n} X(a)$ has fewer than n terms. The primes in $(n/2, n]$ occur in F_n as isolated linear terms. Our main new ingredient is a square-root descent in characteristic two, which shows that once there are at least $\lfloor \sqrt{n} \rfloor$ such primes, a counterexample can be chosen that avoids all of them. More generally, for any polynomial $C + z_1 + \cdots + z_r$ over $\mathbb{F}_2$ we prove this whenever $r \geq \sigma(C)$, where $\sigma(C)$ is the largest number of terms of C in one exponent-parity class. This condition cannot be weakened to $r = \sigma(C) - 1$. For $n \geq 67$ the proof is completed by a splitting inequality for minimal counterexamples and an explicit count of rough-factor components. The same descent settles $n \leq 66$. All estimates are elementary and are proved in full.

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1. INTRODUCTION

1.1. **The conjecture.** For finite sets $A, B \subset \mathbb{N} = \{1, 2, \ldots\}$ we follow [8] and write $A * B$ for the set of integers having an *odd* number of representations $x = ab$ with $a \in A$ and $b \in B$. Put $[n] = \{1, \ldots, n\}$. Then

$$A * [n] = A \triangle 2A \triangle \cdots \triangle nA = \bigtriangleup_{a \in A} \{a, 2a, \ldots, na\},$$

where $\triangle$ denotes symmetric difference and $kA = \{ka : a \in A\}$.

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Theorem A. *For every finite nonempty set $A \subset \mathbb{N}$ and every integer $n \geq 1$,*

$$|A * [n]| \geq n.$$

The bound is attained, for example, by $A = \{a\}$, by $A = \{a, 2a\}$ when n is even, and by $A = [n]$; see [8, Section 1]. In the last case every representation $x = ab$ with $a \neq b$ is paired with $x = ba$, so $[n] * [n] = \{1^2, 2^2, \dots, n^2\}$. Ke and Meyer [10] showed that equality holds for arbitrarily large sets A. So the extremal configurations are far from unique, and a proof must deal with substantial cancellation.

1.2. **History.** The problem comes from coding theory. Pilz [17] studied the minimal distance of certain polynomial near-ring codes (for near-rings see [16]). In our notation he showed that $|A * [n]| \geq n$ for $n \leq 6$. He also reported that a computer search found no set with $|A * [n]| < n$. The resulting question, whether the minimal distance always equals n, became known as Pilz's conjecture. Pach and Szabó [14] explained its equivalence with Theorem A. They proved it for $A = [k]$, and proved the general lower bound $|A * [n]| \gg n/(\log n)^\lambda$ for every $\lambda > \lambda_0 \approx 0.2223$.

Huang, Ke and Pilz [9] independently settled the case $A = [k]$, referred to as the *1-2-3 Conjecture*. They stated the general case as the *Extended 1-2-3 Conjecture*. This conjecture is unrelated to the graph-theoretic 1-2-3 Conjecture.

Ke and Meyer [10] proved the conjecture for $|A| \leq 3$. Nagy and Pach [13] proved the analogous statement for sumsets.

Most recently, Holdridge and Pach [8] proved the conjecture for $n \leq 8$ and for all $n \geq 2(3^{80} - 321)$. They also obtained $|A * [n]| \gg kn/\log k$ when $k = |A|$ is small compared with n, and they studied the limiting density $\lim_n |A * [n]|/n$. Before the present work, the conjecture was therefore open for all $9 \leq n < 2(3^{80} - 321)$.

The conjecture is related to colourings of the n-smooth numbers [2]. In particular, the case of odd $|A|$ would follow from the existence of an n-colouring of $\mathbb{N}$ in which $a, 2a, \dots, na$ receive distinct colours for every a. It is also an arithmetic relative of Pak's problem on the odd area of unit discs [15, 18].

1.3. **Main results.** We prove Theorem A for every n. Throughout, primes are encoded as variables (see Section 2). For $a \in \mathbb{N}$ put

$$X(a) = \prod_p x_p^{v_p(a)}, \qquad F_n = \sum_{a=1}^{n} X(a),$$

where $v_p(a)$ is the exponent of p in a. For a field $\mathbb{K}$ and a nonzero polynomial $P \in \mathbb{K}[x_p : p \text{ prime}]$, let $\mathrm{wt}(P)$ be the number of its nonzero terms, and put

$$\mathrm{dist}_{\mathbb{K}}(P) = \min\{\mathrm{wt}(PH) : H \neq 0\}.$$

We write dist for $\mathrm{dist}_{\mathbb{K}}$ when the field is clear from the context. By unique factorisation, $|A * [n]| = \mathrm{wt}(F_n H_A)$ over $\mathbb{F}_2$, where $H_A = \sum_{a \in A} X(a)$. So Theorem A states that $\mathrm{dist}_{\mathbb{F}_2}(F_n) = n$. Since wt cannot increase under an $\mathbb{F}_2$-linear map applied coefficientwise, the same holds over any field of characteristic two, and for vector-valued multipliers (Corollary 3.9).

The primes $p \in (n/2, n]$ are special. Each divides exactly one element of $[n]$, namely p itself, so it enters F_n as a single linear term. We call these primes *fresh* and write $\mathcal{P}_n$ for the set of them and $r(n) = |\mathcal{P}_n| = \pi(n) - \pi(n/2)$. Thus

$$F_n = C_n + \sum_{p \in \mathcal{P}_n} x_p,$$

where C_n does not involve the fresh variables. The first main ingredient is a general elimination theorem for such polynomials. In set language it reads as follows. Recall that two integers lie in the same *square class* if their ratio is the square of a rational number, that is, if they have the same squarefree part.

Theorem 1.1 (Elimination of fresh primes)**.** *Let $B \subset \mathbb{N}$ be finite and let T be a set of primes. Suppose that each $p \in T$ lies in B and divides no other element of B, and put $B_0 = B \setminus T$. Let $\sigma(B_0)$ be the largest number of elements of B_0 lying in one square class, with $\sigma(\emptyset) = 0$. If $|T| \geq \sigma(B_0)$, then*

$$\min_{A \neq \emptyset} |A * B| = \min_{\substack{A \neq \emptyset \\ \gcd(a,p)=1 \ (a \in A,\ p \in T)}} \big(|A * B_0| + |T|\,|A|\big).$$

The hypothesis $|T| \geq \sigma(B_0)$ cannot be omitted (Example 3.7).

The polynomial version (Theorem 3.6) holds for every polynomial $C + z_1 + \cdots + z_r$ over $\mathbb{F}_2$ with $r \geq \sigma(C)$. For $B = [n]$ the squares form the largest square class, so the hypothesis is $r(n) \geq \lfloor \sqrt{n} \rfloor$, and this holds for every $n \geq 67$ (Proposition 5.1). A counterexample at such n may therefore be assumed to avoid all fresh primes. Each of the $r(n)$ fresh rows then contains a complete copy of the multiplier, so the multiplier has fewer than $n/r(n) \approx 2 \log n$ terms.

The second ingredient is a lower bound for multipliers that involve only a few small primes. It holds over every field.

Theorem 1.2 (Expansion for localised support)**.** *Let $\mathbb{K}$ be any field and let $n \geq 67$. Let $D \in \mathbb{K}[x_p]$ be a nonconstant polynomial with u terms whose terms have no common monomial factor, and let q be the largest prime whose variable occurs in D. If*

$$q \leq n/2 \qquad \text{and} \qquad q < 2(u-1),$$

then $\mathrm{wt}(F_n D) \geq n$.

The two hypotheses are exactly what the reductions deliver. Fresh-prime elimination gives $q \leq n/2$. The condition $q < 2(u-1)$ holds for every normalised counterexample that is minimal under deletion of terms, by a splitting inequality (Lemma 4.1).

1.4. **Strategy.** The proof for $n \geq 67$ consists of three moves (Fig. 1).

(i) *Remove the fresh primes.* The identity $F_n \cdot F_n = F_n^2$, which has exactly n terms, shows that a multiplier may profit from containing fresh primes. In characteristic two, however, any such profit is controlled by squares. Choose a short multiple G of F_n of least total width in the non-fresh variables, and then of least weight. Formal derivatives in the fresh variables synchronise the fresh rows of G. What remains is divisible by F_n^2, and each of its square classes is the square of another multiple of F_n of strictly smaller total width. Minimality makes each such class cost at least n terms. When there are at least $\lfloor \sqrt{n} \rfloor$ fresh primes, no such class can be paid for, and the counterexample can be replaced by one that avoids the fresh primes (Section 3).

(ii) *Confine the remaining primes.* Delete terms of the multiplier while it remains a counterexample. Split the resulting multiplier at the first positive power of a prime q. Both pieces then produce at least n terms, and they can cancel only through integers divisible by q. This forces $q < 2(u-1)$, where u is the number of terms (Section 4).

(iii) *Count rough components.* A multiplier supported on the primes up to q cannot change the q-rough part of an integer. So the product splits into disjoint pieces, one for each q-rough $b \leq n$. Lower bounds for the pieces, combined with elementary sieve and Chebyshev estimates, give at least n terms (Sections 6 and 7).

The threshold 67 is exactly where the counting hypothesis of the first move becomes available: $r(66) = 7 < 8 = \lfloor \sqrt{66} \rfloor$, while $r(n) \geq \lfloor \sqrt{n} \rfloor$ for all $n \geq 67$. For $n \leq 66$ we apply the same descent directly with target n. Explicit lower bounds for the cores C_n show that any correction would be too expensive (Section 8). Within that range, only the cases $n \leq 10$ are used as inputs to other cases.

1.5. **Relation to previous work.** Several ingredients go back to Holdridge and Pach [8], and we indicate them where they are used.

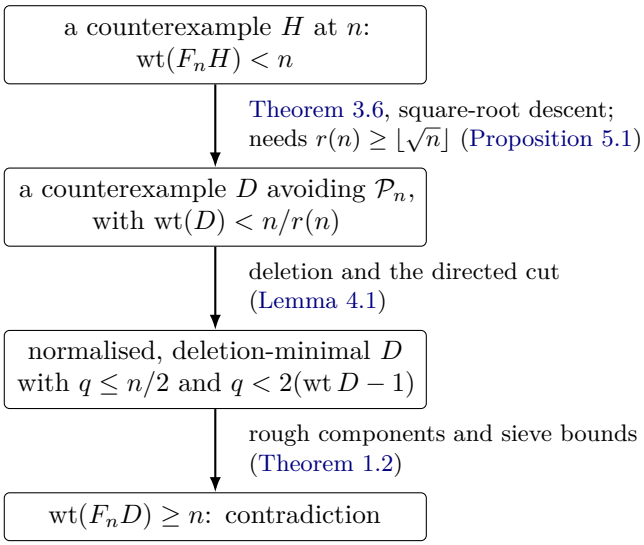


FIGURE 1. The proof of Theorem A for a fixed $n \geq 67$. No case $n' \neq n$ is used.

- The ring structure of finite sets under $\triangle$ and $*$ and its identification with $\mathbb{F}_2[x_p]$ are in [8, Lemmas 2.1–2.2]. Polynomials over $\mathbb{F}_2$ also appear in the coding-theoretic origin of the problem [17, 14] and in the proof of [13].
- The decomposition by rough parts is [8, Lemma 2.4] (for n-rough parts). Slicing a multiplier by p-adic valuation is [8, Section 3].
- The fact that the primes in $(n/2, n]$ appear exactly once in $[n]$, and the importance of this fact, are emphasised in [8, Section 1]. Holdridge and Pach remove these primes one at a time. They expand the remaining multiplier in powers of $[n] \setminus \mathcal{Q}$ and use the identity $S * S = \{s^2 : s \in S\}$ [8, Lemma 6.2 and Section 6]. They also observe that, for products with $[n] \setminus T$, one may assume that no element of A is divisible by a prime of T [8, proof of Lemma 6.1]. This is a precursor of Corollary 3.8. Our square-root descent (Section 3) is an exact, threshold version of this mechanism. It removes all fresh primes at once, with no loss, as soon as there are $\lfloor\sqrt{n}\rfloor$ of them.
- The splitting inequality of Lemma 4.1 is a variant of [8, Lemma 4.1 and Corollary 4.3], combined here with minimality under deletion.
- The lower bound through rough numbers in $(n/2, n]$ for smooth multipliers is [8, Lemma 4.4]. Theorem 1.2 is an explicit refinement valid for every $n \geq 67$.
- The overall shape of the second and third moves follows Holdridge and Pach's proofs of their Theorem 1.4 and of the case $k \geq 81$ of their Lemma 6.1. If a large prime divides an element of A, they split A using [8, Corollary 4.3]. Otherwise A is smooth, and they count rough numbers in $(n/2, n]$ [8, Lemma 4.4].
- The identity $S * S = \{s^2 : s \in S\}$ is [8, Lemma 6.2]. The pairing idea behind it goes back to the proof of [10, Theorem 5.1].
- The multiplicativity of weight parity is [8, Lemma 3.4], generalising [17, Lemma 4].
- The two- and three-term cases of Theorem A are due to Ke and Meyer [10]. We reprove them in our framework because the small cases use them.
- The example $\{1, 2\} * \{1, 2, 4\} = \{1, 8\}$, showing that Theorem A fails for general B in place of $[n]$, is from [8].

Holdridge and Pach used the explicit rough-number estimates of Fan and of Fan–Pomerance [5, 6] and the prime estimates of Rosser–Schoenfeld [19]. To keep the present paper self-contained and suitable for formal verification, we instead prove the weaker estimates we need from first principles. These are Chebyshev's argument [3] with the binomial-coefficient bound familiar

from Erdős's proof of Bertrand's postulate [4] (see also [7, 12]), and a Legendre-type sieve for rough numbers in the sense of Buchstab [1].

1.6. **Further results.** Besides Theorem A, Theorem 1.1 and Theorem 1.2, the proof gives several statements of independent interest.

- Theorem A holds over every field of characteristic two and for vector-valued multipliers (Corollary 3.9).
- The polynomial elimination theorem (Theorem 3.6) comes with a lower bound for every residual square class (Proposition 3.4).
- Theorem A holds for every multiplier with at most four terms, extending the case $|A| \le 3$ of [10], and for every multiplier supported on at most three primes (Corollaries 8.7 and 8.10 and Theorem 8.9). These are proved early, using only the cases $n \le 10$.
- Over any field, minimal counterexamples have a rigid structure: every element of their normalised support divides $(2u-3)!$, where u is the number of terms (Proposition 9.2).

1.7. **Organisation and notation.** Section 2 sets up the polynomial model, the endpoint bound and the rough decomposition. Section 3 proves the elimination theorem, and Section 4 the splitting inequality. Section 5 assembles the proof for $n \ge 67$ from two propositions, proved in Section 6 (counting fresh primes) and Section 7 (counting rough components). Section 8 treats $n \le 66$, and Section 9 discusses sharpness and open problems. Appendix A records the logical dependencies, which we hope will be useful for formalisation.

Throughout, p and q denote primes, $\pi(t)$ is the number of primes at most t (for real t), and log is the natural logarithm. Table 1 collects the main notation.

Table 1. Main notation.

$X(a)$, F_n	monomial of $a \in \mathbb{N}$; $F_n = \sum_{a \le n} X(a)$
wt, supp, $\mathrm{dist}_{\mathbb{K}}$	number of terms; support; minimal weight of a nonzero multiple
$\mathcal{P}_n$, $r(n)$	fresh primes $p \in (n/2, n]$; $r(n) = \lvert\mathcal{P}_n\rvert = \pi(n) - \pi(n/2)$
C_n	F_n with the fresh terms removed
$P_{[\epsilon]}$, $\sigma(P)$	full parity class of P; largest size of a parity class
$F_j^{(q)}$	sum of $X(a)$ over q-smooth $a \le j$
$\Phi_q(t)$, $I_q(t)$	number of q-rough $b \le t$; number of q-rough $b \in (t/2, t]$
$\varrho(q)$, δ_q	$\prod_{p \le q}(1 - 1/p)$; discrepancy bound for Φ_q
width_x, W	$\deg_x - \mathrm{ord}_x$; total width in the base variables

2. The polynomial model

2.1. **Encoding.** Let $\mathbb{K}$ be a field. We work in the polynomial ring $\mathbb{K}[x_p : p \text{ prime}]$ in one variable for each prime. Every polynomial involves only finitely many variables. Unique factorisation shows that $a \mapsto X(a)$ is a bijection from $\mathbb{N}$ onto the monomials, with $X(ab) = X(a)X(b)$. For finite $A, B \subset \mathbb{N}$ we therefore have, over $\mathbb{F}_2$,

$$\Big(\sum_{a \in A} X(a)\Big)\Big(\sum_{b \in B} X(b)\Big) = \sum_{x \in A * B} X(x), \tag{2.1}$$

since the coefficient of $X(x)$ on the left is the number of representations $x = ab$, reduced modulo 2. This is the identification of [8, Lemmas 2.1 and 2.2]. In particular $|A * [n]| = \mathrm{wt}(F_n H_A)$ with $H_A = \sum_{a \in A} X(a)$, and Theorem A is equivalent to $\mathrm{dist}_{\mathbb{F}_2}(F_n) = n$. The upper bound $\mathrm{dist}_{\mathbb{F}_2}(F_n) \le n$ is witnessed by $H = 1$.

For example, with $x = x_2$, $y = x_3$, $z = x_5$,

$$(1 + x)(1 + x + y + x^2 + z + xy) = 1 + y + z + x^3 + xz + x^2 y,$$

which records $\{1, 2\} * [6] = \{1, 3, 5, 8, 10, 12\}$.

We write $\operatorname{supp} P$ for the set of monomials of P with nonzero coefficient. A nonzero polynomial is *normalised* if its terms have no common monomial factor, that is, if its minimum exponent in every variable is zero. For a variable x, write $\deg_x P$ and $\operatorname{ord}_x P$ for the largest and smallest exponent of x in P, and put $\operatorname{width}_x(P) = \deg_x P - \operatorname{ord}_x P$. Widths add under multiplication, since lowest and highest terms multiply.

Lemma 2.1 (Elementary reductions)**.** *Let $P \neq 0$.*

(a) *Multiplying P or a multiplier by a monomial changes neither weights nor $\operatorname{dist}_{\mathbb{K}}(P)$. In particular, multipliers may be assumed normalised.*
(b) *If P involves only the variables $\mathbf{x}$, then $\operatorname{dist}_{\mathbb{K}}(P)$ is attained by a multiplier in $\mathbb{K}[\mathbf{x}]$.*
(c) *A nonzero multiple of a non-monomial has at least two terms.*

Proof. (a) is clear. For (b), let $H \neq 0$ be any multiplier and expand $H = \sum_\beta \mathbf{w}^\beta H_\beta$ in the remaining variables $\mathbf{w}$, with $H_\beta \in \mathbb{K}[\mathbf{x}]$. Then $PH = \sum_\beta \mathbf{w}^\beta (PH_\beta)$, and the summands have disjoint supports. Hence $\operatorname{wt}(PH_\beta) \leq \operatorname{wt}(PH)$ for a nonzero H_β. For (c), a divisor of a monomial is a monomial. □

2.2. **Endpoints and parity.** The following bound is a polynomial form of the slicing by p-adic valuation used in [8, Section 3].

Lemma 2.2 (Endpoint bound)**.** *Let y be a variable and $P = \sum_{i=0}^{d} a_i y^i$ with $d \geq 1$, $a_0 a_d \neq 0$ and every a_i independent of y. Then over any field*

$$\operatorname{dist}(P) \geq \operatorname{dist}(a_0) + \operatorname{dist}(a_d).$$

Consequently, if P contains a nonzero constant term and a positive pure power of each of s distinct variables, then $\operatorname{dist}(P) \geq s + 1$. In particular $\operatorname{dist}(z_1 + \cdots + z_r) \geq r$ and $\operatorname{dist}(1 + z_1 + \cdots + z_r) \geq r + 1$.

Proof. Let $H \neq 0$, and let h_e and h_f be the lowest and highest nonzero y-coefficients of H. The lowest and highest y-coefficients of PH are $a_0 h_e$ and $a_d h_f$. Both are nonzero, and they lie in different y-degrees because $d \geq 1$. This proves the first claim.

For the second, slice in one of the s variables. The lowest coefficient still contains the constant and the other $s - 1$ pure powers. The highest coefficient is nonzero, so it has distance at least one. Induction on s completes the proof.

For $z_1 + \cdots + z_r$ the case $r = 1$ is trivial. For $r \geq 2$, slice in z_1: the lowest coefficient is $z_2 + \cdots + z_r$ and the highest is 1, so induction on r applies. The bound for $1 + z_1 + \cdots + z_r$ is the second claim with $s = r$. □

Lemma 2.3 (Weight parity [8, Lemma 3.4])**.** *Over $\mathbb{F}_2$, $\operatorname{wt}(PH) \equiv \operatorname{wt}(P)\operatorname{wt}(H) \pmod 2$. In particular every multiple of a polynomial of even weight has even weight.*

The special case that $|A * [n]|$ is even for even n is [17, Lemma 4].

Proof. Over $\mathbb{F}_2$, $\operatorname{wt}(P) \bmod 2$ is the value of P at $(1, 1, \dots)$, and evaluation is a ring homomorphism. □

2.3. **Parity classes.** Over $\mathbb{F}_2$, the map $P \mapsto P^2$ is additive, and it sends distinct monomials to distinct monomials. Hence $\operatorname{wt}(P^2) = \operatorname{wt}(P)$. Every $P \in \mathbb{F}_2[\mathbf{x}]$ has a unique decomposition

$$P = \sum_{\epsilon \in \{0,1\}^{\mathbf{x}}} \mathbf{x}^\epsilon P_\epsilon^2, \tag{2.2}$$

obtained by grouping monomials according to the parities of their exponents. We call $P_{[\epsilon]} = \mathbf{x}^\epsilon P_\epsilon^2$ the *full parity class* of P indexed by ϵ. Distinct classes have disjoint supports, and $\operatorname{wt}(P_{[\epsilon]}) = \operatorname{wt}(P_\epsilon)$. Let $\sigma(P) = \max_\epsilon \operatorname{wt}(P_{[\epsilon]})$. In integer language, two monomials $X(a)$ and

$X(b)$ lie in the same class exactly when a and b have the same squarefree part. Since every integer is uniquely a squarefree number times a square,

$$F_n = \sum_{\substack{a \le n \\ a \text{ squarefree}}} X(a)\, F^2_{\lfloor \sqrt{n/a} \rfloor}. \tag{2.3}$$

Thus the class of F_n indexed by a squarefree a has $\lfloor \sqrt{n/a} \rfloor$ terms. Hence $\sigma(F_n) = \lfloor \sqrt{n} \rfloor$, attained by the squares, and every other class has at most $\lfloor \sqrt{n/2} \rfloor$ terms. The identity $F_n \cdot F_n = F_n^2$ is the polynomial form of $[n] * [n] = \{k^2 : k \le n\}$ [10, 8].

A polynomial all of whose exponents of a variable z are even has formal derivative $\partial_z P = 0$, and conversely. In characteristic two, $\partial_z(z^k) = k z^{k-1}$ is nonzero exactly when k is odd. So ∂_z maps each monomial that is odd in z to a distinct monomial even in z, and kills the rest. Every derivative $\partial_z P$ is even in z.

2.4. **Rough components.** Let T be a finite set of primes. An integer is *T-smooth* if all its prime factors lie in T, and *T-rough* if none does. For real $t \ge 0$ and integers $j \ge 1$ put

$$F_j^{(T)} = \sum_{\substack{a \le j \\ a\ T\text{-smooth}}} X(a), \qquad \Phi_T(t) = \#\{b \le t :\ b\ T\text{-rough}\}, \qquad I_T(t) = \Phi_T(t) - \Phi_T(t/2).$$

Thus $I_T(t)$ counts the T-rough integers in $(t/2, t]$. When T is the set of primes at most q, we say *q-smooth* and *q-rough*, and write $F_j^{(q)}$, Φ_q and I_q.

Each $a \le n$ factors uniquely as $a = bc$ with b T-rough and c T-smooth. A multiplier D involving only primes in T preserves the T-rough part of every monomial. Hence (compare [8, Lemma 2.4])

$$F_n D = \sum_{\substack{b \le n \\ b\ T\text{-rough}}} X(b)\, F^{(T)}_{\lfloor n/b \rfloor} D, \qquad \mathrm{wt}(F_n D) = \sum_{\substack{b \le n \\ b\ T\text{-rough}}} \mathrm{wt}(F^{(T)}_{\lfloor n/b \rfloor} D), \tag{2.4}$$

over any field. The summands have pairwise disjoint supports. We call $F^{(T)}_{\lfloor n/b \rfloor} D$ the *component* of b, and $\lfloor n/b \rfloor$ its *index*. For each $k \ge 1$ there are exactly $\Phi_T(n/k)$ components of index at least k. There are $I_T(n)$ components of index one, and each of them is a copy of D.

Lemma 2.4 (Component profile)**.** *Let D be a normalised nonconstant polynomial over any field, and let q be the largest prime whose variable occurs in D. Then for every $j \ge 1$,*

$$\mathrm{wt}(F_j^{(q)} D) \ \ge \ \begin{cases} \mathrm{wt}(D) & j = 1, \\ 2(1 + \pi(j)) & 2 \le j < q, \\ 1 + \pi(q) & j \ge q. \end{cases}$$

Proof. For $j = 1$ the component is D. If $2 \le j < q$, then $F_j^{(q)} = F_j$ is independent of x_q. Since D is normalised and involves x_q, it has at least two nonzero x_q-slices. Multiplication by F_j keeps these slices in distinct x_q-degrees. Each product contains a nonzero multiple of F_j, and F_j contains 1 and the variables x_p for $p \le j$, so each has at least $1 + \pi(j)$ terms by Lemma 2.2. If $j \ge q$, then $F_j^{(q)}$ contains 1 and x_p for every $p \le q$, and Lemma 2.2 gives $1 + \pi(q)$. □

Summing the profile over components gives two budgets that we use repeatedly. Every component of index at least two has at least four terms if $q \ge 5$. Moreover, the profile values at indices $1, 2, 3, 4$ and at every index ≥ 5 are at least $\mathrm{wt}(D), 4, 6, 6, 8$ when $q \ge 17$, because $\pi(17) = 7$. Writing $u = \mathrm{wt}(D)$, we obtain

$$\mathrm{wt}(F_n D) \ge 4\Phi_q(n) + (u - 4) I_q(n) \qquad (q \ge 5), \tag{2.5}$$

$$\mathrm{wt}(F_n D) \ge u\, I_q(n) + 4\Phi_q(n/2) + 2\Phi_q(n/3) + 2\Phi_q(n/5) \qquad (q \ge 17). \tag{2.6}$$

Inequality (2.5) holds for every $u \ge 1$, since u is the exact weight of each index-one component. The lower bound through the index-one components alone, $\mathrm{wt}(F_n D) \ge u\, I_q(n)$, is the bound of [8, Lemma 4.4].

3. Eliminating the fresh primes

All polynomials in this section have coefficients in $\mathbb{F}_2$.

3.1. **Setting.** Fix base variables $\mathbf{x} = (x_1, \dots, x_t)$ and fresh variables $\mathbf{z} = (z_1, \dots, z_r)$ with $r \geq 1$. Let $0 \neq C \in \mathbb{F}_2[\mathbf{x}]$ and put

$$Z = z_1 + \cdots + z_r, \qquad F = C + Z.$$

By Lemma 2.1(b) we may compute $\operatorname{dist}(F)$ inside $\mathbb{F}_2[\mathbf{x}, \mathbf{z}]$. A polynomial is *fresh-even* if it lies in $\mathbb{F}_2[\mathbf{x}, z_1^2, \dots, z_r^2]$. Only base variables enter the total width

$$W(P) = \sum_{i=1}^{t} \operatorname{width}_{x_i}(P) \qquad (P \neq 0).$$

For a variable x_i, the width of F equals $\deg_{x_i} C$, because Z has x_i-exponent 0. Hence

$$\operatorname{width}_{x_i}(F) = \deg_{x_i} C \qquad (1 \leq i \leq t). \tag{3.1}$$

Fix an integer target N with $1 \leq N \leq \operatorname{wt}(C) + r$. Call a nonzero multiple G of F *short* if $\operatorname{wt}(G) < N$.

Definition 3.1 (Extremal multiples)**.** A short multiple G of F is *extremal* if it minimises W among all short multiples, and minimises wt among the short multiples with that value of W.

Extremal multiples exist whenever short ones do, because W and wt take values in $\mathbb{N} \cup \{0\}$. The descent consists of replacing an extremal multiple by others that are still extremal, until it takes a rigid form.

Lemma 3.2 (Basic facts)**.** (a) *F is a prime element of $\mathbb{F}_2[\mathbf{x}, \mathbf{z}]$. It divides no monomial, it does not divide Z, and it does not divide $z_i + z_j$ for $i \neq j$.*

(b) *If $\operatorname{wt}(C) = 1$, there are no short multiples. Hence, if short multiples exist, then C is not a monomial and $W(F) \geq 1$.*

(c) (Stability.) *Let G be extremal and let G' be a nonzero multiple of F with $\operatorname{wt}(G') \leq \operatorname{wt}(G)$ and $\operatorname{width}_{x_i}(G') \leq \operatorname{width}_{x_i}(G)$ for all i. Then G' is extremal, with the same weight and the same widths as G.*

(d) *If G is extremal and $\mathbf{z}^\beta$ is a monomial in the fresh variables, then $\mathbf{z}^\beta G$ is extremal.*

(e) *For $i \neq j$ the derivation $\partial_{ij} = \partial_{z_i} + \partial_{z_j}$ satisfies $\partial_{ij} F = 0$. Hence $\partial_{ij}(FH) = F\,\partial_{ij} H$, and ∂_{ij} maps multiples of F to multiples of F. Every ∂_{z_i} maps a polynomial to one whose base exponents occur among those of the original.*

Proof. (a) Since F is linear in z_1 with coefficient 1, substituting $z_1 = C + z_2 + \cdots + z_r$ identifies $\mathbb{F}_2[\mathbf{x}, \mathbf{z}]/(F)$ with the domain $\mathbb{F}_2[\mathbf{x}, z_2, \dots, z_r]$. So F is prime, and F divides a polynomial exactly when that polynomial vanishes under the substitution. A monomial maps to a nonzero product, since $C + z_2 + \cdots + z_r \neq 0$. The polynomial Z maps to $C \neq 0$. For $i, j \neq 1$, $z_i + z_j$ maps to itself. Finally $z_1 + z_j$ maps to $C + \sum_{k \neq 1, j} z_k$, which is nonzero because $C \neq 0$ involves no fresh variable.

(b) If $C = \mathbf{x}^\alpha$ is a monomial, then $\operatorname{dist}(F) \geq r + 1 = \operatorname{wt}(C) + r \geq N$. This follows by slicing $F = \mathbf{x}^\alpha + z_1 + \cdots + z_r$ successively in $z_1, \dots, z_r$ as in Lemma 2.2. Otherwise C involves some x_i, and $W(F) \geq \deg_{x_i} C \geq 1$ by (3.1).

(c) G' is short, so $W(G') \geq W(G)$ by extremality. Since each width of G' is at most the corresponding width of G, all widths are equal. Extremality then gives $\operatorname{wt}(G') \geq \operatorname{wt}(G)$, hence equality.

(d) Multiplying by $\mathbf{z}^\beta$ changes neither base widths nor weight.

(e) $\partial_{z_i} F = 1$ for every i. The rest is the Leibniz rule and the fact that ∂_{z_i} does not change base exponents. □

Note that $\operatorname{wt}(\partial_{z_i} G)$ equals the number $m_i(G)$ of terms of G whose z_i-exponent is odd.

3.2. **Synchronising the fresh rows.**

Proposition 3.3 (Synchronised normal form)**.** *Suppose short multiples of F exist. Then there is an extremal multiple G, of weight m say, with*

$$G = G_0 + ZD, \qquad m = \mathrm{wt}(G_0) + r\,u, \qquad u := \mathrm{wt}(D) \le \tfrac{m}{2},$$

where G_0 and D are fresh-even and nonzero. In particular $ru < m < N$, and $\partial_{z_i} G = D$ for every i.

Proof. By Lemma 3.2(b), $W(F) \ge 1$. Start from any extremal G, of weight m, and write $m_i = m_i(G)$.

Step 1 (parity normalisation). Multiplying by z_i replaces m_i by $m - m_i$ and leaves the other m_j unchanged. By Lemma 3.2(d) we may therefore assume $m_i \le m/2$ for all i.

Step 2 (synchronisation). Suppose that $m_i < m/2$ for some i, and let $j \ne i$. By Lemma 3.2(e), $\partial_{ij} G$ is a multiple of F of weight at most $m_i + m_j < m$, with no larger base widths than G. By Lemma 3.2(c) it must vanish. Hence all the derivatives $\partial_{z_j} G$ coincide; call the common value D. Since $D = \partial_{z_j} G$ is even in z_j for every j, it is fresh-even. Put $G_0 = G + ZD$. Then

$$\partial_{z_j} G_0 = D + (\partial_{z_j} Z)D + Z\,\partial_{z_j} D = D + D + 0 = 0$$

for every j, so G_0 is fresh-even. In $G = G_0 + ZD$ the summands G_0, $z_1 D, \dots, z_r D$ have pairwise disjoint supports: G_0 is even in every fresh variable, while $z_j D$ is odd in z_j only. Hence $m = \mathrm{wt}(G_0) + r\ \mathrm{wt}(D)$ and $u = \mathrm{wt}(D) = m_i \le m/2$. If $r = 1$, Step 2 applies without any condition on m_1: it is just the decomposition by z_1-parity.

Step 3 (the balanced case). Suppose $r \ge 2$ and $m_i = m/2$ for every i. Write $y = z_1$, $y' = z_2$ and

$$G = B_{00} + yB_{10} + y'B_{01} + yy'B_{11},$$

with each B_{kl} even in y and y'. The conditions $m_1 = m_2 = m/2$ give $\mathrm{wt}(B_{10}) = \mathrm{wt}(B_{01}) =: b$ and $\mathrm{wt}(B_{00}) = \mathrm{wt}(B_{11}) =: h$ with $m = 2(h + b)$.

If $b > 0$, put $G' = \partial_{12} G = B_{10} + B_{01} + (y + y')B_{11}$. This is a multiple of F of weight at most $2b + 2h = m$, with no larger base widths. It is nonzero. Indeed, $G' = 0$ would force $B_{11} = 0$, hence $h = 0$ and $B_{00} = 0$, and $B_{10} = B_{01}$, so $G = (y + y')B_{10}$. Then $F \mid B_{10}$ by Lemma 3.2(a), and B_{10} is a nonzero multiple of weight $m/2 < m$ with no larger widths, contradicting Lemma 3.2(c). So G' is extremal by Lemma 3.2(c), and its number of terms odd in y is $\mathrm{wt}(B_{11}) = h < m/2$.

If $b = 0$, put $G'' = \partial_{12}(yG) = B_{00} + y^2 B_{11}$. Here yG is extremal by Lemma 3.2(d). If $G'' = 0$ then $G = y(y + y')B_{11}$, and the same argument gives a contradiction. So G'' is extremal, and it has no terms odd in y.

In both cases we obtain an extremal multiple with $m_1 < m/2$. Multiplying by suitable z_j with $j \ge 3$ restores $m_j \le m/2$ for those j without changing m_1, and Step 2 applies.

Step 4 ($D \ne 0$). Let G be as produced by Step 2, and suppose $D = 0$, so that every fresh derivative of G vanishes. Write $G = FH$. Then $0 = \partial_{z_1} G = H + F\,\partial_{z_1} H$, so $F \mid H$ and $G = F^2 E$. Since F^2 is fresh-even and $\partial_{z_j} G = F^2 \partial_{z_j} E$, the polynomial E is fresh-even. Decompose it as $E = \sum_\epsilon \mathbf{x}^\epsilon E_\epsilon^2$ over base parities. Then

$$G = \sum_\epsilon \mathbf{x}^\epsilon (F E_\epsilon)^2,$$

and the summands have disjoint supports. Choose ϵ with $E_\epsilon \ne 0$. Then FE_ϵ is a nonzero multiple of F of weight $\mathrm{wt}(\mathbf{x}^\epsilon (FE_\epsilon)^2) \le m < N$. Its widths satisfy $2\,\mathrm{width}_{x_i}(FE_\epsilon) = \mathrm{width}_{x_i}(\mathbf{x}^\epsilon (FE_\epsilon)^2) \le \mathrm{width}_{x_i}(G)$. Since $W(G) \ge W(F) \ge 1$, we get $W(FE_\epsilon) \le W(G)/2 < W(G)$, contradicting extremality.

Step 5 ($G_0 \ne 0$). If $r = 1$, then $m = \mathrm{wt}(G_0) + u$ and $u \le m/2$ force $\mathrm{wt}(G_0) \ge m/2 > 0$. If $r \ge 2$ and $G_0 = 0$, then $G = ZD$. Since F is prime and does not divide Z, we get $F \mid D$. But $D = \partial_{z_1} G$ is nonzero, has weight $m/r < m$, and has no larger widths than G, contradicting Lemma 3.2(c).

Finally $G_0 \ne 0$ gives $ru < m$. □

3.3. **The residual and its square classes.**

Proposition 3.4 (Residual classes cost the full target)**.** *Let $G = G_0 + ZD$ be as in Proposition 3.3, and put*

$$R = G + FD = G_0 + CD.$$

Then R is fresh-even and divisible by F^2, and every nonzero full parity class of R has at least N terms.

Proof. The two expressions for R agree because $ZD + ZD = 0$. The second shows that R is fresh-even. Write $G = FH$. Then $D = \partial_{z_1} G = H + F\,\partial_{z_1} H$, so $R = FH + FH + F^2\partial_{z_1} H = F^2 J$ with $J = \partial_{z_1} H$. Since R is fresh-even and $\partial_{z_j} R = F^2 \partial_{z_j} J$, the polynomial J is fresh-even too.

If $R = 0$, the assertion is immediate; assume $R \neq 0$.

The cofactor H has positive width in some base variable. Otherwise $H = \mathbf{x}^\beta H'$ with $H' \in \mathbb{F}_2[\mathbf{z}]$. Let $c_0 \in \{0, 1\}$ be the constant term of C. Then $FH' = (C - c_0)H' + (c_0 + Z)H'$. The terms of $(C - c_0)H'$ are copies of H' in distinct nonzero base monomials, and $(c_0 + Z)H'$ lies in base degree zero. By Lemma 2.2,

$$\operatorname{wt}(FH') = (\operatorname{wt}(C) - c_0)\operatorname{wt}(H') + \operatorname{wt}((c_0 + Z)H') \geq \operatorname{wt}(C) - c_0 + r + c_0 \geq N,$$

contradicting $\operatorname{wt}(FH') = \operatorname{wt}(G) < N$.

Width of the residual. Fix a base variable x. Every exponent of x in G_0 or in $D = \partial_{z_1} G$ is an exponent of x in G. Since $R = G_0 + CD$, we get $\operatorname{ord}_x R \geq \operatorname{ord}_x G$ and $\deg_x R \leq \deg_x G + \deg_x C$. Using (3.1) and $\operatorname{width}_x(G) = \operatorname{width}_x(F) + \operatorname{width}_x(H)$, this gives

$$\operatorname{width}_x(R) \leq \operatorname{width}_x(G) + \operatorname{width}_x(F) = 2\operatorname{width}_x(G) - \operatorname{width}_x(H). \tag{3.2}$$

Square classes. Since $R = F^2 J$ is fresh-even, decomposing J by full parity shows that the nonzero full parity classes of R are $R_{[\epsilon]} = \mathbf{x}^\epsilon (FJ_\epsilon)^2$ for base parity vectors ϵ and polynomials J_ϵ. Here $\operatorname{wt}(R_{[\epsilon]}) = \operatorname{wt}(FJ_\epsilon)$, and by (3.2)

$$\operatorname{width}_x(FJ_\epsilon) = \tfrac{1}{2}\operatorname{width}_x(R_{[\epsilon]}) \leq \tfrac{1}{2}\operatorname{width}_x(R) \leq \operatorname{width}_x(G) - \tfrac{1}{2}\operatorname{width}_x(H)$$

for every base variable x, with strict inequality for at least one x. Thus FJ_ϵ is a nonzero multiple of F with $W(FJ_\epsilon) < W(G)$. By extremality it is not short, so $\operatorname{wt}(R_{[\epsilon]}) = \operatorname{wt}(FJ_\epsilon) \geq N$. □

The polynomial G differs from the multiple FD only by R. The fresh-odd part of FD is ZD and its fresh-even part is CD. A residual class can therefore be cancelled only by the corresponding class of CD. This yields the following comparison. Let a denote the number of nonzero full parity classes of R.

Corollary 3.5 (Protected classes)**.** *In the situation of Proposition 3.4:*

(i) *If $\operatorname{wt}((CD)_{[\epsilon]}) < ru + N - m$, then $R_{[\epsilon]} = 0$. In particular this holds whenever $\operatorname{wt}((CD)_{[\epsilon]}) \leq ru$.*

(ii) *If every full parity class of CD has at most M terms, then*

$$m \geq \operatorname{wt}(FD) + a(N - 2M) \qquad \text{and} \qquad a(N - M) \leq m - ru.$$

(iii) *Let M bound every class of CD. If either $N \geq 2M$ and $\operatorname{wt}(FD) \geq N$, or $a \leq 1$ and $\operatorname{wt}(FD) \geq \max\{N, 2M\}$, then $m \geq N$, which is impossible. In applications this shows that no short multiple exists.*

Proof. Classwise, $G_{[\epsilon]} = R_{[\epsilon]} + (FD)_{[\epsilon]}$. Every class of R is fresh-even, and for fresh-even ϵ we have $(FD)_{[\epsilon]} = (CD)_{[\epsilon]}$. Moreover $G_{[\epsilon]}$ is then contained in G_0, whose weight is $m - ru$. If $R_{[\epsilon]} \neq 0$, then

$$m - ru \geq \operatorname{wt}(G_{[\epsilon]}) \geq N - \operatorname{wt}((CD)_{[\epsilon]}),$$

which proves (i). For (ii), each ϵ with $R_{[\epsilon]} \neq 0$ satisfies

$$\operatorname{wt}(G_{[\epsilon]}) \geq N - M \geq \operatorname{wt}((FD)_{[\epsilon]}) + N - 2M.$$

The other classes satisfy $\operatorname{wt}(G_{[\epsilon]}) = \operatorname{wt}((FD)_{[\epsilon]})$. Summing over all classes gives the first inequality. The a classes with $R_{[\epsilon]} \neq 0$ lie in G_0, which gives the second. For (iii): in the first case

$m \geq \mathrm{wt}(FD) \geq N$. In the second, $m \geq \mathrm{wt}(FD) \geq N$ if $a = 0$, and $m \geq \mathrm{wt}(FD)+N-2M \geq N$ if $a = 1$. □

3.4. **The elimination theorem.**

Theorem 3.6 (Fresh-variable elimination)**.** *If $r \geq \sigma(C)$, then*

$$\mathrm{dist}(C + z_1 + \cdots + z_r) = \min_{0 \neq D \in \mathbb{F}_2[\mathbf{x}]} \big(\mathrm{wt}(CD) + r\ \mathrm{wt}(D)\big).$$

In particular, the minimum on the left is attained by a multiplier involving no fresh variable.

Proof. Let d denote the right-hand side. A multiplier $D \in \mathbb{F}_2[\mathbf{x}]$ gives $FD = CD + z_1 D + \cdots + z_r D$ with disjoint summands, so $\mathrm{dist}(F) \leq d \leq \mathrm{wt}(C) + r$ (take $D = 1$). Suppose $\mathrm{dist}(F) < d$. Then short multiples exist for the target $N = d$, and Proposition 3.3 provides $G = G_0 + ZD$ with fresh-even $D \neq 0$ of weight u.

Every full parity class of CD is a sum of products $C_{[\epsilon']} \cdot D_{[\epsilon'']}$ with $\epsilon' + \epsilon'' \equiv \epsilon$. For fixed ϵ, each class of D occurs once, so the class has at most $\sigma(C)\, u \leq ru$ terms. By Corollary 3.5(i), $R = 0$, that is, $G = FD$.

Write $D = \sum_\alpha \mathbf{z}^{2\alpha} D_\alpha$ with $D_\alpha \in \mathbb{F}_2[\mathbf{x}]$. Then

$$G = FD = \sum_\alpha \Big(\mathbf{z}^{2\alpha} C D_\alpha + \sum_{i=1}^{r} \mathbf{z}^{2\alpha} z_i D_\alpha \Big).$$

All these summands have pairwise disjoint supports, since their fresh exponent vectors are 2α and $2\alpha + e_i$. Hence

$$m = \mathrm{wt}(G) = \sum_\alpha \big(\mathrm{wt}(CD_\alpha) + r\ \mathrm{wt}(D_\alpha)\big) \geq d,$$

because some $D_\alpha \neq 0$. This contradicts $m < N = d$. □

Proof of Theorem 1.1. Let $\mathbf{x}$ be the variables of the primes dividing elements of B_0, and let $z_p = x_p$ for $p \in T$. By hypothesis $\sum_{b \in B} X(b) = C + \sum_{p \in T} z_p$ with $C = \sum_{b \in B_0} X(b) \in \mathbb{F}_2[\mathbf{x}]$. Two elements of B_0 lie in the same square class exactly when their monomials lie in the same full parity class, so $\sigma(C) = \sigma(B_0)$.

If $B_0 = \emptyset$, so that $B = T$, both sides equal $|T|$ by Lemma 2.2. So let $B_0 \neq \emptyset$; then $|T| \geq \sigma(B_0) \geq 1$. By (2.1), the left-hand side of the theorem is $\mathrm{dist}(C + \sum_{p \in T} z_p)$, which may be computed in $\mathbb{F}_2[\mathbf{x}, \mathbf{z}]$ by Lemma 2.1(b). By Theorem 3.6 it equals the minimum of $\mathrm{wt}(CD) + |T|\,\mathrm{wt}(D)$ over nonzero $D \in \mathbb{F}_2[\mathbf{x}]$.

On the right-hand side, A may involve further primes outside T. Let $\mathbf{w}$ be the corresponding variables and let D be free of the z_p. Expanding $D = \sum_\beta \mathbf{w}^\beta D_\beta$ with $D_\beta \in \mathbb{F}_2[\mathbf{x}]$ gives $\mathrm{wt}(CD) + |T|\,\mathrm{wt}(D) = \sum_\beta \big(\mathrm{wt}(CD_\beta) + |T|\,\mathrm{wt}(D_\beta)\big)$. So such primes never lower the minimum. Finally, for A with no element divisible by a prime of T, the encoding gives $\mathrm{wt}(CH_A) = |A * B_0|$ and $\mathrm{wt}(H_A) = |A|$. □

Example 3.7 (The hypothesis $r \geq \sigma(C)$ is needed)**.** Let $x = x_2$, $z = x_3$ and $C = (1+x)^5 = 1 + x + x^4 + x^5$, so that $\sigma(C) = 2$ and $r = 1$. In integers, $B = \{1, 2, 3, 16, 32\}$, $T = \{3\}$ and $B_0 = \{1, 2, 16, 32\}$. Then

$$G = (1+x)z^3 + 1 + x^{16} = (C + z)\,(1+x)(z^2 + Cz + C^2)$$

has four terms. With the corresponding A,

$$A = \{1, 2, 3, 4, 8, 9, 12, 18, 48, 192, 256, 512, 1024, 2048\},$$

we get $A * B = \{1, 27, 54, 65536\}$. On the other hand, every $D \in \mathbb{F}_2[x] \setminus \{0\}$ has $\mathrm{wt}(CD) + \mathrm{wt}(D) \geq 5$. For one-term D the value is 5, and for D with at least three terms it is at least $2+3$. For two terms we may take $D = 1 + x^j$ with $j \geq 1$; then the exponent set $\{0, 1, 4, 5\}$ of C meets its translate by j in at most two points, so $\mathrm{wt}(CD) \geq 4$. As in the proof of Theorem 1.1, further primes cannot lower this minimum, so $|A * B_0| + |A| \geq 5$ for every nonempty A coprime to 3. So the conclusion of Theorem 1.1 fails for $B = \{1, 2, 3, 16, 32\}$.

The multiple G also shows the structure described in Propositions 3.3 and 3.4. With target $N = 5$, it has the synchronised form with $m = 4$, $D = (1 + x)z^2$, $u = 2$, $G_0 = 1 + x^{16}$ and $R = F^2(1 + x)^6$. The residual has a single parity class, of weight $6 \geq N$, as Proposition 3.4 requires of extremal multiples. (We have not proved that G is extremal.) The matching class of CD has weight $4 > ru + N - m = 3$, so it is not protected, and the cancellation is genuine.

3.5. **Application to intervals.** For F_n the fresh variables are x_p with $p \in \mathcal{P}_n$, and $F_n = C_n + Z_n$ with $Z_n = \sum_{p\in\mathcal{P}_n} x_p$. Removing these primes from $[n]$ leaves every square. So, by (2.3), $\sigma(C_n) = \lfloor\sqrt{n}\rfloor$.

Corollary 3.8 (Counterexamples avoid the fresh primes)**.** *Let $n \geq 2$ with $r(n) \geq \lfloor\sqrt{n}\rfloor$. If $\operatorname{dist}_{\mathbb{F}_2}(F_n) < n$, then there is a nonzero D involving only the primes at most $n/2$ with*

$$\operatorname{wt}(F_nD) = \operatorname{wt}(C_nD) + r(n)\operatorname{wt}(D) < n.$$

In particular $\operatorname{wt}(D) < n/r(n)$.

Proof. By Lemma 2.1(b) we may work with the variables of F_n. These are the base variables x_p with $p \leq n/2$ and the fresh variables. Now apply Theorem 3.6. □

Corollary 3.9 (Characteristic two and vector coefficients)**.** *Let $\mathbb{K}$ be a field of characteristic two, or let V be a vector space over $\mathbb{F}_2$. If Theorem A holds for a given n, then $\operatorname{wt}(F_nH) \geq n$ for every nonzero finitely supported $H = \sum_a h_aX(a)$ with coefficients in $\mathbb{K}$, respectively in V. Here weight counts monomials with nonzero coefficient.*

Proof. The coefficients of H span a finite-dimensional $\mathbb{F}_2$-space. Choose an $\mathbb{F}_2$-linear functional λ on it that does not vanish on all of them. Applying λ coefficientwise commutes with multiplication by F_n, because F_n has coefficients in $\mathbb{F}_2$. It also cannot enlarge supports. So $\operatorname{wt}(F_nH) \geq \operatorname{wt}(F_n\lambda(H)) \geq n$. □

4. Minimal counterexamples and the directed cut

Fix $n \geq 1$ and a field $\mathbb{K}$. A nonzero $H \in \mathbb{K}[x_p]$ is a *counterexample at n* if $\operatorname{wt}(F_nH) < n$. It is *minimal under deletion* if no polynomial obtained by deleting some, but not all, of its terms is a counterexample. Every counterexample contains one that is minimal under deletion: keep deleting terms while a counterexample remains. Deletion never introduces a new variable, so it preserves any restriction on the primes involved. Dividing by the greatest common monomial factor also preserves minimality under deletion, by Lemma 2.1(a).

The next lemma is a variant of the splitting inequality of Holdridge and Pach [8, Lemma 4.1 and Corollary 4.3]. Here the split is taken at a prime power, intersections are counted from the lower side only, and the input is minimality under deletion.

Lemma 4.1 (Directed cut)**.** *Let D be a normalised counterexample at n that is minimal under deletion, with u terms, and let q be a prime. For $a \geq 0$ let u_a be the number of terms of D whose x_q-exponent equals a. Then for every $b > 0$ with $u_b > 0$,*

$$2\sum_{a<b} u_a \left\lfloor \frac{n}{q^{b-a}} \right\rfloor > n. \tag{4.1}$$

Consequently every prime q whose variable occurs in D satisfies

$$q < 2u_0 \leq 2(u - 1).$$

Proof. Split $D = L + U$, where L consists of the terms with x_q-exponent less than b. Both pieces are nonzero (the exponent 0 is occupied because D is normalised), so both are proper. Minimality gives $\operatorname{wt}(F_nL) \geq n$ and $\operatorname{wt}(F_nU) \geq n$. Over any field,

$$\operatorname{wt}(P + Q) \geq \operatorname{wt}(P) + \operatorname{wt}(Q) - 2\,|\operatorname{supp} P \cap \operatorname{supp} Q|,$$

so $n > \operatorname{wt}(F_nD)$ forces $|\operatorname{supp}(F_nL) \cap \operatorname{supp}(F_nU)| > n/2$.

Every monomial of F_nU has x_q-exponent at least b. A monomial of F_nL with x_q-exponent at least b arises from a term of L with exponent $a < b$ and an integer $j \leq n$ with $q^{b-a} \mid j$.

There are $\lfloor n/q^{b-a} \rfloor$ such j. Hence the intersection has at most $\sum_{a<b} u_a \lfloor n/q^{b-a} \rfloor$ elements, proving (4.1).

If x_q occurs in D, let b be its least positive exponent. Then only $a = 0$ contributes, and $2u_0\lfloor n/q^b \rfloor > n$ gives $q \le q^b < 2u_0$. Finally $u_0 \le u - 1$ because the exponent b is occupied. □

A more general form of this inequality, which counts all gaps between occupied exponents, is given in Section 9. It shows that the support of a minimal counterexample consists of divisors of $(2u-3)!$.

5. Proof of Theorem A for $n \ge 67$

The proof for $n \ge 67$ uses two counting statements, proved in the next two sections.

Proposition 5.1 (Fresh primes)**.** *For every integer $n \ge 67$ we have $r(n) \ge \lfloor \sqrt{n} \rfloor$. Moreover:*

(a) *for $67 \le n \le 1023$, $r(n) \ge 8$ and*

$$\beta(n) := 10\, r(n) + 4\pi(n/2) + 2\pi(n/3) + 2\pi(n/5) - 48 \;\ge\; n;$$

(b) *for $1024 \le n \le 32768$, $r(n) > n/30$;*

(c) *for $n \ge 32768$, $r(n) > 7n/(20 \log n)$.*

The value 67 is optimal for the first assertion: $r(66) = 7 < 8 = \lfloor \sqrt{66} \rfloor$.

The second statement is Theorem 1.2 from the introduction. We prove it in Section 7. It holds over every field, and it assumes no minimality of D.

Proof of Theorem A for $n \ge 67$. Fix $n \ge 67$ and suppose that $\operatorname{dist}_{\mathbb{F}_2}(F_n) < n$. By Proposition 5.1, $r(n) \ge \lfloor \sqrt{n} \rfloor$. So Corollary 3.8 provides a counterexample at n involving only primes at most $n/2$. Delete terms until it is minimal under deletion, and divide by its greatest common monomial factor. Call the result D, with u terms. Both operations preserve the restriction to primes at most $n/2$ and the property of being a counterexample.

A monomial multiplier gives exactly n terms, so D is not a monomial. Let q be the largest prime whose variable occurs in D. Then $q \le n/2$, and Lemma 4.1 gives $q < 2(u-1)$. Now Theorem 1.2 gives $\operatorname{wt}(F_n D) \ge n$, contradicting the choice of D. □

Remark 5.2. The argument works at a single fixed n. It uses no instance of Theorem A at any other interval length, and in particular no induction on n.

6. Counting the fresh primes

This section proves Proposition 5.1. For $n \le 1023$ we compute directly from the residues modulo 30. For larger n we use an elementary form of Chebyshev's method.

6.1. From 67 to 100.

Lemma 6.1. *For $67 \le n \le 100$ we have $r(n) \ge 8 + \mathbf{1}_{n \ge 73} + \mathbf{1}_{n \ge 97} \ge \lfloor \sqrt{n} \rfloor$ and $\beta(n) \ge 104 \ge n$.*

Proof. An integer in $(33, 100]$ coprime to 30 is prime unless it is divisible by 7, since every composite number up to 100 has a prime factor at most 7. The integers in $(33, 100]$ coprime to 30 are

$$37,\ 41,\ 43,\ 47,\ 49,\ 53,\ 59,\ 61,\ 67,\ 71,\ 73,\ 77,\ 79,\ 83,\ 89,\ 91,\ 97,$$

and removing 49, 77 and 91 leaves the primes. At $n = 67$ the fresh primes are $37, 41, 43, 47, 53, 59, 61, 67$, so $r(67) = 8$. As n runs from 67 to 100:

- a prime enters $(n/2, n]$ at $n = 71, 73, 79, 83, 89, 97$;
- a prime leaves it at $n = 74, 82, 86, 94$.

Pair the arrivals $71, 79, 83, 89$ with the departures $74, 82, 86, 94$, respectively. Each arrival precedes its departure, and 73 and 97 remain unpaired. This proves the bound for $r(n)$. The comparison with $\lfloor \sqrt{n} \rfloor$ holds because $\lfloor \sqrt{n} \rfloor \le 8$ for $n \le 80$, $\lfloor \sqrt{n} \rfloor = 9$ for $81 \le n \le 99$, and $\lfloor \sqrt{100} \rfloor = 10$.

The primes up to 33 are $2, 3, \dots, 31$, eleven in number. So $\pi(n/2) \geq \pi(33) = 11$, $\pi(n/3) \geq \pi(22) = 8$ and $\pi(n/5) \geq \pi(13) = 6$ throughout the range. Hence $\beta(n) \geq 80+44+16+12-48 = 104$. □

6.2. **From 101 to 1023.** Let $\mathcal{R} = \{1, 7, 11, 13, 17, 19, 23, 29\}$ be the residues coprime to 30. For real $x \geq 0$ write $\lfloor x \rfloor = 30h + s$ with $0 \leq s < 30$, and put

$$N_{30}(x) = 8h + \#\{a \in \mathcal{R} : a \leq s\},$$

the number of positive integers at most x coprime to 30. Let $\Phi_7(x) = N_{30}(x) - N_{30}(x/7)$ count the 7-rough integers up to x.

Lemma 6.2 (An exact prime count below 1024)**.** *Let* $(p_1, \dots, p_7) = (11, 13, 17, 19, 23, 29, 31)$. *For every real* x *with* $7 \leq x \leq 1023$,

$$\pi(x) = 3 + \Phi_7(x) - \sum_{i=1}^{7} \max\{\Phi_7(x/p_i) - i,\, 0\}.$$

Proof. A 7-rough integer up to 1023 is 1, a prime greater than 7, or a product of two primes greater than 7, because $11^3 = 1331 > 1023$. In a product pp' with $p \leq p'$ we have $p \leq \sqrt{1023} < 32$, so $p = p_i$ for some i. For fixed p_i, the admissible p' are the primes in $[p_i, x/p_i]$. Since $x/p_i \leq 93 < 121 = 11^2$, every 7-rough integer up to x/p_i is 1 or a prime. Exactly i of them, namely $1, p_1, \dots, p_{i-1}$, lie below p_i. So there are $\max\{\Phi_7(x/p_i) - i, 0\}$ such products. Each composite is counted once, prime squares included. Adding the primes $2, 3, 5, 7$ and removing the integer 1 gives the formula. □

Proposition 6.3. *For* $101 \leq n \leq 1023$ *we have* $r(n) \geq \lfloor \sqrt{n} \rfloor$ *and* $\beta(n) \geq n$.

Proof. Cover $[101, 1023]$ by the thirteen intervals $[a, b]$ of Table 2. For $n \in [a, b]$, monotonicity gives

$$r(n) \geq r_0 := \pi(a) - \pi(b/2), \qquad \beta(n) \geq \beta_0 := 10r_0 + 4\pi(a/2) + 2\pi(a/3) + 2\pi(a/5) - 48.$$

It therefore suffices that $r_0 \geq \lfloor \sqrt{b} \rfloor$ and $\beta_0 \geq b$ in each row, which the table shows. All prime counts follow from Lemma 6.2.

For instance, in the row $[422, 524]$, the five arguments 422, 262, 211, 140, 84 (integer parts of a, $b/2$, $a/2$, $a/3$, $a/5$) give the following.

- N_{30} values: $113, 70, 57, 38, 23$; at $x/7$: $16, 10, 8, 6, 3$.
- Φ_7 values: $97, 60, 49, 32, 20$.
- Correction sums: $18, 8, 5, 1, 0$. For $x = 422$ the seven corrections are $8, 6, 3, 1, 0, 0, 0$.
- Prime counts: $82, 55, 47, 34, 23$.

Hence $r_0 = 27 \geq 22$ and $\beta_0 = 270 + 188 + 68 + 46 - 48 = 524 = b$. □

6.3. **An elementary Chebyshev estimate.** Let $\vartheta(x) = \sum_{p \leq x} \log p$ and $\psi(x) = \sum_{p^k \leq x} \log p$. We use Chebyshev's combination of factorials [3] together with the bound $\operatorname{lcm}(1, \dots, k) < 4^k$, which goes back to the central-binomial argument in Erdős's proof of Bertrand's postulate [4]; see also [7, 12]. Put

$$\mu_1 = \frac{14\log 2 + 9\log 3 + 5\log 5}{30}, \qquad \mu_2 = \frac{6}{5}\mu_1, \qquad \eta(x) = \frac{5}{4\log 6}\Big(\log x + \tfrac{1}{2}\log 6 - \tfrac{6}{5}\Big)^2 + 9.$$

Lemma 6.4 (Elementary Chebyshev bounds)**.** *For* $x \geq 30$,

$$\mu_1 x - \tfrac{5}{2}\log x + 2 < \psi(x) < \mu_2 x + \eta(x),$$

and for $x \geq 256$,

$$0 \leq \psi(x) - \vartheta(x) < \tfrac{8}{3}\sqrt{x}.$$

Consequently, for $n \geq 256$,

$$r(n)\log n > \tfrac{2}{5}\mu_1 n - \tfrac{8}{3}\sqrt{n} - \tfrac{5}{2}\log n + 2 - \eta(n/2). \tag{6.1}$$

TABLE 2. Prime counts for $101 \le n \le 1023$.

a	b	$\pi(a)$	$\pi(b/2)$	$\pi(a/2)$	$\pi(a/3)$	$\pi(a/5)$	r_0	$\lfloor\sqrt{b}\rfloor$	β_0
101	117	26	16	15	11	8	10	10	150
118	141	30	19	17	12	9	11	11	172
142	165	34	22	20	15	9	12	12	200
166	195	38	25	23	16	11	13	13	228
196	225	44	29	25	18	12	15	15	262
226	273	48	32	30	21	14	16	16	302
274	345	58	39	33	24	16	19	18	354
346	421	68	46	40	30	19	22	20	430
422	524	82	55	47	34	23	27	22	524
525	636	99	66	55	40	27	33	25	636
637	765	115	75	66	47	31	40	27	772
766	913	135	87	76	54	36	48	30	916
914	1023	156	97	88	62	42	59	31	1102

Proof. Write $\lambda(t) = \log\lfloor t\rfloor!$ and $h(t) = t\log t - t$.

Factorials. We first show, for $t \ge 1$,

$$h(t) - \tfrac{1}{2}\log t + \tfrac{1+\log 2}{2} \le \lambda(t) \le h(t) + \tfrac{1}{2}\log t + 1. \tag{6.2}$$

For an integer $k \ge 1$, the trapezoidal rule underestimates $\int_1^k \log v\,dv = h(k) + 1$ by concavity, so $\lambda(k) \le h(k) + \frac{1}{2}\log k + 1$. This extends to real t since h and log increase on $[1,\infty)$. For the lower bound, concavity gives $\log i \ge \int_{i-1/2}^{i+1/2} \log v\,dv$. Summing over $i \le k = \lfloor t\rfloor$ gives

$$\lambda(t) \ge h(k+\tfrac{1}{2}) - h(\tfrac{1}{2}) = h(k+\tfrac{1}{2}) + \tfrac{1+\log 2}{2}.$$

Finally $h(t) - h(k+\frac{1}{2}) \le \frac{1}{2}\log t$, because $h' = \log$ and $t < k+1$.

Chebyshev's combination. Put

$$\Xi(x) = \lambda(x) - \lambda(x/2) - \lambda(x/3) - \lambda(x/5) + \lambda(x/30).$$

In the combination of the functions $h(x/d)$, the coefficients of $x\log x$ and of x vanish. The remaining linear term is $\mu_1 x$. Applying (6.2) to each term, for $x \ge 30$,

$$\mu_1 x - \tfrac{5}{2}\log x + \log 60 - 2 \;\le\; \Xi(x) \;\le\; \mu_1 x + \tfrac{5}{2}\log x - \log 30 + \tfrac{1-3\log 2}{2}. \tag{6.3}$$

The numerical bounds below give $\log 60 - 2 > 2$ and $-\log 30 + \frac{1-3\log 2}{2} < -3$.

By Legendre's formula $\lambda(x) = \sum_{p^k \le x} \lfloor x/p^k\rfloor \log p$, we have

$$\Xi(x) = \sum_{p^k \le x} f(x/p^k)\log p, \qquad f(t) = \lfloor t\rfloor - \lfloor t/2\rfloor - \lfloor t/3\rfloor - \lfloor t/5\rfloor + \lfloor t/30\rfloor.$$

The function f has period 30 and is constant on each interval $[k, k+1)$. In one period it equals 1 on the intervals with left endpoint

$$1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 14, 17, 19, 23, 29,$$

and 0 elsewhere. In particular $0 \le f \le 1$, and $f = 1$ on $[1, 6)$. Hence

$$\psi(x) - \psi(x/6) \le \Xi(x) \le \psi(x),$$

and the lower bound for ψ follows from (6.3).

The bound $\psi(t) < t\log 4$ for $t \ge 1$. Equivalently, $\operatorname{lcm}(1,\dots,k) < 4^k$ for every integer $k \ge 1$. We induct on k; the case $k = 1$ is clear. Passing from $k-1$ to an even k multiplies the lcm by at most 2, because the only prime power that can be even is a power of 2. For $k = 2m+1$,

every prime power $p^e \in (m+1, 2m+1]$ with $p^{e-1} \le m+1$ contributes a single factor p. That p divides $\binom{2m+1}{m}$, by Legendre's formula at the level p^e. Hence

$$\operatorname{lcm}(1,\dots,2m+1) \le \operatorname{lcm}(1,\dots,m+1)\binom{2m+1}{m} < 4^{m+1}4^m,$$

using $2\binom{2m+1}{m} \le 2^{2m+1}$.

Upper bound for ψ. Iterate $\psi(x) \le \Xi(x) + \psi(x/6)$ until $t = x/6^k \in [5, 30)$. Each argument $x/6^j$ with $j < k$ is at least 30, so (6.3) applies. The linear terms sum to $\mu_2(x-t)$. Moreover $\psi(t) < t\log 4$ and $(\log 4 - \mu_2)t < 9$. The logarithmic terms sum to

$$\sum_{j=0}^{k-1}\Big(\tfrac{5}{2}(\log x - j\log 6) - 3\Big) = -\tfrac{5}{4}\log 6\, k^2 + \Big(\tfrac{5}{2}\log x + \tfrac{5}{4}\log 6 - 3\Big)k \le \eta(x) - 9,$$

by completing the square.

Prime powers. Grouping even exponents,

$$\psi(x) - \vartheta(x) = \psi(\sqrt{x}) + \sum_{\substack{a\ge 3\\ a \text{ odd}}} \vartheta(x^{1/a}).$$

At most $\frac{\log x}{2\log 2} - \frac{3}{2}$ odd exponents $a \ge 5$ contribute. With $\psi(t) < t\log 4$ this gives

$$\frac{\psi(x) - \vartheta(x)}{\sqrt{x}} < \log 4\Big(1 + x^{-1/6} + (\tfrac{\log x}{2\log 2} - \tfrac{3}{2})x^{-3/10}\Big).$$

For $x \ge 256$ we have $x^{-1/6} < \frac{2}{5}$. The last term decreases for $x \ge 256$, because $\log 2 > \frac{2}{3}$, and at $x = 256$ it equals $\frac{5}{2}\cdot 2^{-12/5} < \frac{1}{2}$. Since $\log 4 < \frac{7}{5}$, the bound is less than $\frac{7}{5}\cdot\frac{19}{10} < \frac{8}{3}$.

Conclusion. Every prime in $(n/2, n]$ contributes at most $\log n$ to $\vartheta(n) - \vartheta(n/2)$, so $r(n)\log n \ge \vartheta(n) - \vartheta(n/2)$. Insert the bounds above, noting that $\mu_1 n - \frac{1}{2}\mu_2 n = \frac{2}{5}\mu_1 n$.

Numerical constants. All logarithm bounds used come from the series $\log\frac{1+t}{1-t} = 2\sum_{j\ge 0} t^{2j+1}/(2j+1)$.

- The first three terms at $t = \frac{1}{3}$ give $\log 2 > \frac{693}{1000}$. Bounding the remaining denominators below by 3 gives $\log 2 < \frac{25}{36} < \frac{7}{10}$.
- The first four terms at $t = \frac{1}{2}$ give $\log 3 > \frac{1098}{1000}$.
- The first two terms at $t = \frac{1}{9}$ give $\log\frac{5}{4} > \frac{223}{1000}$, hence $\log 5 > \frac{1609}{1000}$.
- The same tail bound at $t = \frac{1}{5}$ gives $\log\frac{3}{2} < \frac{73}{180}$.

These give $\mu_1 > \frac{23}{25}$ and $\log 6 > \frac{25}{14}$, so $\frac{5}{4\log 6} < \frac{7}{10}$. They also give $\log 2 + \frac{6}{5} - \frac{1}{2}\log 6 = \frac{6}{5} - \frac{1}{2}\log\frac{3}{2} > \frac{99}{100}$, as well as the two constants used for (6.3). □

6.4. **Large n.**

Lemma 6.5. *We have $r(n) > n/30$ for $1024 \le n \le 32768$ and $r(n) > 7n/(20\log n)$ for $n \ge 32768$. In particular $r(n) > \sqrt{n}$ for all $n \ge 1024$.*

Proof. Put $L = \log n$ and

$$w(L) = \tfrac{7}{10}(L - \tfrac{99}{100})^2 + \tfrac{5}{2}L + 7, \qquad g(L) = \tfrac{46}{125} - \tfrac{8}{3}e^{-L/2} - e^{-L}w(L).$$

For $L > 6$, the constants at the end of the proof of Lemma 6.4 give

$$\eta(n/2) = \tfrac{5}{4\log 6}\Big(L - (\log 2 + \tfrac{6}{5} - \tfrac{1}{2}\log 6)\Big)^2 + 9 < \tfrac{7}{10}(L - \tfrac{99}{100})^2 + 9.$$

Together with $\mu_1 > \frac{23}{25}$, inequality (6.1) then yields $r(n)L/n > g(L)$.

For $L \ge 6$, g is increasing and strictly concave. Indeed $g' = \frac{4}{3}e^{-L/2} + e^{-L}(w - w')$ and $g'' = -\frac{2}{3}e^{-L/2} - e^{-L}(w - 2w' + w'')$, and both

$$w - w' = \tfrac{7}{10}\Big\{(L - \tfrac{99}{100})^2 - 2(L - \tfrac{99}{100})\Big\} + \tfrac{5}{2}L + \tfrac{9}{2}$$

and

$$w - 2w' + w'' = \tfrac{7}{10}\Big\{\big(L - \tfrac{299}{100}\big)^2 - 2\Big\} + \tfrac{5}{2}L + 2$$

are positive there. Hence $g(L) - L/30$ is strictly concave as well.

The endpoint $n = 1024$. Here $L < 7$, $\sqrt{n} = 32$ and $w(7) < 50$, so

$$g(L) - \tfrac{L}{30} > \tfrac{46}{125} - \tfrac{1}{12} - \tfrac{50}{1024} - \tfrac{7}{30} = \tfrac{481}{192000} > 0.$$

The endpoint $n = 32768 = 2^{15}$. Here $L < \frac{21}{2}$, $\sqrt{n} > 180$ and $w(\frac{21}{2}) < 97$, so

$$g(L) - \tfrac{7}{20} > \tfrac{46}{125} - \tfrac{2}{135} - \tfrac{97}{32768} - \tfrac{7}{20} = \tfrac{24881}{110592000} > 0.$$

At $n = 32768$ also $L/30 < \frac{7}{20}$. By concavity, $g(L) > L/30$ on the whole first range, which gives $r(n) > n/30$. Since g is increasing, $g(L) > \frac{7}{20}$ on the second range, which gives $r(n) > 7n/(20 \log n)$.

Finally, $n/30 > \sqrt{n}$ for $n \geq 1024$. The function $\sqrt{n}/\log n$ increases for $n \geq e^2$, and $\frac{7}{20} \cdot \frac{180}{21/2} = 6 > 1$, so $7n/(20 \log n) > \sqrt{n}$ for $n \geq 32768$. □

Proof of Proposition 5.1. Combine Lemma 6.1, Proposition 6.3 and Lemma 6.5. The bound $r(n) \geq 8$ for $n \leq 1023$ follows from the first two. □

7. Counting rough components

This section proves Theorem 1.2. Throughout, $\mathbb{K}$ is any field and $D \in \mathbb{K}[x_p]$ is normalised and nonconstant, with u terms and largest prime q. All estimates rest on (2.4), Lemma 2.4 and the budgets (2.5)–(2.6), in which every coefficient is nonnegative (for (2.5), as soon as $u \geq 4$, which holds whenever we use it). So lower bounds for the counts Φ_q and I_q may be substituted directly.

7.1. Sieve estimates.

Put $\varrho(q) = \prod_{p \leq q}(1 - 1/p)$.

Lemma 7.1 (Discrepancy)**.** *For every real $x \geq 0$, $|\Phi_q(x) - \varrho(q)x| \leq \delta_q$, where*

$$\delta_2 = \tfrac{1}{2}, \qquad \delta_3 = \tfrac{2}{3}, \qquad \delta_q = \tfrac{14}{15}\, 2^{\pi(q)-3} \quad (q \geq 5).$$

Consequently $I_q(x) \geq \varrho(q)x/2 - 2\delta_q$.

Proof. The cases $q = 2, 3$ follow from the residues modulo 2 and 6. Modulo 30 the rough residues are $\mathcal{R} = \{1, 7, 11, 13, 17, 19, 23, 29\}$. At these points the values of $15\Phi_5(x) - 4x$ are

$$11,\ 2,\ 1,\ 8,\ 7,\ 14,\ 13,\ 4.$$

Between consecutive jumps the error $\Phi_5(x) - \frac{4}{15}x$ decreases linearly. Just before the jump at a listed point it equals the listed value divided by 15, minus one. The error is periodic modulo 30, so $|\Phi_5(x) - \frac{4}{15}x| \leq \frac{14}{15}$. Adding a prime p to the sieve gives $\Phi_{T\cup\{p\}}(x) = \Phi_T(x) - \Phi_T(x/p)$. This multiplies the density by $1 - 1/p$ and at most doubles the discrepancy. Induction proves the claim, and subtracting the estimates at x and $x/2$ gives the bound for I_q. □

Lemma 7.2 (A one-step sieve)**.** *Let T and T' be disjoint finite sets of primes, and suppose $|\Phi_T(x) - \varrho x| \leq \delta$ for all $x \geq 0$. Put*

$$\alpha = \varrho\Big(1 - \sum_{p \in T'} \frac{1}{p}\Big), \qquad \bar{\delta} = (1 + |T'|)\,\delta.$$

Then $\Phi_{T\cup T'}(x) \geq \alpha x - \bar{\delta}$ and $I_{T\cup T'}(x) \geq \alpha x/2 - 2\bar{\delta}$. If moreover $|I_T(x) - \varrho x/2| \leq \delta'$ for all x, then $I_{T\cup T'}(x) \geq \alpha x/2 - (1 + |T'|)\delta'$.

Proof. Remove from the T-rough integers the multiples of each $p \in T'$, charging an integer once for each such divisor. Since multiplication by $p \notin T$ preserves T-roughness, this union bound gives

$$\Phi_{T\cup T'}(x) \geq \Phi_T(x) - \sum_{p \in T'} \Phi_T(x/p).$$

The same argument on $(x/2, x]$ gives $I_{T\cup T'}(x) \geq I_T(x) - \sum_{p\in T'} I_T(x/p)$. Insert the discrepancy bounds, using $\delta' = 2\delta$ when no better bound is available. □

Lemma 7.3 (Half-interval discrepancy)**.** *For every real $x \geq 0$, $|I_5(x) - \frac{2}{15}x| \leq \frac{22}{15}$.*

Proof. The error $I_5(x) - \frac{2}{15}x$ has period 60, since $I_5(x+60) = I_5(x) + 8$. Between jumps it has slope $-\frac{2}{15}$. It jumps up at the integers coprime to 30 and down at twice those integers. At the sixteen upward jumps $t \in (0, 60)$ the values of $15I_5(t) - 2t$ are:

t	1	7	11	13	17	19	23	29
$15I_5(t) - 2t$	13	1	8	19	11	22	14	2
t	31	37	41	43	47	49	53	59
$15I_5(t) - 2t$	13	1	−7	4	−4	7	14	2

So the error is at most $\frac{22}{15}$. For nonintegral $x \in (0, 60)$, the reflection $b \mapsto 60 - b$ of residues gives $I_5(60 - x) = 8 - I_5(x)$, so the error changes sign. Every value is a one-sided limit of values at nonintegral points, which proves the lower bound. □

For a prime $q \geq 7$ put

$$\alpha_q = \tfrac{4}{15}\Big(1 - \sum_{7\leq p\leq q} \tfrac{1}{p}\Big), \qquad \nu_q = \pi(q) - 2.$$

Lemmas 7.1 to 7.3, applied with $T = \{2, 3, 5\}$ and $|T'| = \pi(q) - 3$, give

$$\Phi_q(x) \geq \alpha_q x - \tfrac{14}{15}\nu_q, \qquad I_q(x) \geq \tfrac{1}{2}\alpha_q x - \tfrac{22}{15}\nu_q. \tag{7.1}$$

7.2. **Small support primes.**

Proposition 7.4. *Theorem 1.2 holds when $q \leq 13$.*

Proof. Let $n \geq 67$. The hypothesis $q < 2(u-1)$ gives $u \geq 3, 4, 5, 7, 8$ for $q = 3, 5, 7, 11, 13$, respectively.

$q = 2$. Every component has at least two terms (Lemma 2.4), so $\mathrm{wt}(F_n D) \geq 2\Phi_2(n) = 2\lceil n/2\rceil \geq n$.

$q = 3$. The profile is $u \geq 3$ at index one, 4 at index two (as $2 < q$), and 3 beyond. Hence

$$\mathrm{wt}(F_n D) \geq 3\Phi_3(n) + \big(\Phi_3(n/2) - \Phi_3(n/3)\big).$$

The residues $1, 5$ modulo 6 give $3\Phi_3(n) \geq n - 1$. Consecutive such residues are at most 4 apart, so the interval $(n/3, n/2]$, of length $n/6 \geq 4$, contains a 3-rough integer.

$q = 5$. Every component has at least four terms, so

$$\mathrm{wt}(F_n D) \geq 4\Phi_5(n) \geq \tfrac{16n-56}{15} \geq n \qquad (n \geq 56).$$

$q = 7, 11, 13$. With the least admissible u, the profile from index two onwards is bounded below by

$$(4, 5, 5, \dots), \qquad (4, 6, 6, \dots), \qquad (4, 6, 6, 7, 7, \dots),$$

respectively. Hence $\mathrm{wt}(F_n D) \geq \Lambda_q(n)$, where

$$\begin{aligned}
\Lambda_7(n) &= 5\Phi_7(n) - \Phi_7(n/2) + \Phi_7(n/3) = 4\Phi_7(n) + I_7(n) + \Phi_7(n/3),\\
\Lambda_{11}(n) &= 7\Phi_{11}(n) - 3\Phi_{11}(n/2) + 2\Phi_{11}(n/3) = 4\Phi_{11}(n) + 3I_{11}(n) + 2\Phi_{11}(n/3),\\
\Lambda_{13}(n) &= 8\Phi_{13}(n) - 4\Phi_{13}(n/2) + 2\Phi_{13}(n/3) + \Phi_{13}(n/5)\\
&= 4\Phi_{13}(n) + 4I_{13}(n) + 2\Phi_{13}(n/3) + \Phi_{13}(n/5).
\end{aligned}$$

Larger values of u only add $(u - u_{\min})I_q(n) \geq 0$. Substitute into the second, nonnegative forms the bounds $\Phi_q(x) \geq \alpha_q x - \frac{14}{15}\nu_q$ from (7.1) and $I_q(x) \geq \frac{1}{2}\alpha_q x - 2\cdot\frac{14}{15}\nu_q$ from Lemma 7.2, with the constants

q	7	11	13
α_q	8/35	236/1155	184/1001
$\frac{14}{15}\nu_q$	28/15	14/5	56/15

For these three primes the cruder error $2 \cdot \frac{14}{15}\nu_q$ in the bound for I_q is sufficient, so we use it rather than $\frac{22}{15}\nu_q$ from (7.1). This gives

$$\Lambda_7(n) \geq \tfrac{116n-1372}{105}, \qquad \Lambda_{11}(n) \geq \tfrac{4366n-116424}{3465}, \qquad \Lambda_{13}(n) \geq \tfrac{18952n-840840}{15015}.$$

Since weights are integers, these give $\mathrm{wt}(F_nD) \geq n$ as soon as the right side exceeds $n-1$. This happens for $n \geq 116$, 126, 210 respectively, the conditions being $11n > 1267$, $901n > 112959$ and $3937n > 825825$.

For the remaining ranges we use Table 3. For a row $[a, b]$ put $b_7 = \min(b, 115)$, $b_{11} = \min(b, 125)$ and $b_{13} = b$. Since Φ_q is nondecreasing, $\Lambda_q(n) \geq \Lambda_q^-$ for $a \leq n \leq b_q$. Here Λ_q^- is obtained from the first form of Λ_q by replacing every positive argument n/k by a/k, and the sole negative argument $n/2$ by $b_q/2$. Every entry is at least the corresponding b_q. A dash means that the linear bound already covers the row.

TABLE 3. Lower bounds Λ_q^- for $67 \leq n \leq 209$.

$[a, b]$	Λ_7^-	Λ_{11}^-	Λ_{13}^-
$[67, 76]$	76	89	91
$[77, 85]$	86	102	105
$[86, 94]$	94	110	114
$[95, 101]$	101	121	127
$[102, 110]$	110	132	139
$[111, 145]$	126	149	149
$[146, 181]$	–	–	181
$[182, 209]$	–	–	238

The counts in Table 3 come from N_{30} (Section 6) by inclusion–exclusion over the primes $7, 11, 13$. For $0 \leq t \leq 209$,

$$\begin{aligned}
\Phi_7(t) &= N_{30}(t) - N_{30}(t/7), \\
\Phi_{11}(t) &= N_{30}(t) - N_{30}(t/7) - N_{30}(t/11) + \mathbf{1}_{t \geq 77}, \\
\Phi_{13}(t) &= N_{30}(t) - N_{30}(t/7) - N_{30}(t/11) - N_{30}(t/13) + \mathbf{1}_{t \geq 77} + \mathbf{1}_{t \geq 91} + \mathbf{1}_{t \geq 143}.
\end{aligned}$$

These hold because $t/(pp') < 7$ for distinct $p, p' \in \{7, 11, 13\}$, so that $N_{30}(t/(pp')) = \mathbf{1}_{t \geq pp'}$, and because $7 \cdot 11 \cdot 13 = 1001 > 209$. For example, in the row $[111, 145]$ the values of Φ_q at 111, at $b_q/2$, at 37 and at $111/5$ are $(26, 13, 9, 5)$, $(25, 14, 8, 4)$ and $(24, 15, 7, 3)$ for $q = 7, 11, 13$. The three bounds are therefore $130 - 13 + 9 = 126$, $175 - 42 + 16 = 149$ and $192 - 60 + 14 + 3 = 149$. □

7.3. **Larger support primes up to $n = 1023$.**

Lemma 7.5 (Prime budget)**.** *Suppose $r(n) \geq 8$, $17 \leq q \leq n/2$ and $q < 2(u-1)$. Then*

$$\mathrm{wt}(F_nD) \geq \beta(n) + (u-10)(r(n)-8) \geq \beta(n),$$

with $\beta(n)$ as in Proposition 5.1.

Proof. Every prime in $(n/2, n]$ exceeds q, so $I_q(n) \geq r(n)$. Also $\Phi_q(t) \geq 1 + \pi(t) - \pi(q)$ for all t: when the right side is positive, it counts 1 and the primes in $(q, t]$. Inserting these into (2.6) gives

$$\mathrm{wt}(F_nD) \geq u\, r(n) + 4\pi(n/2) + 2\pi(n/3) + 2\pi(n/5) + 8 - 8\pi(q).$$

Now $q \leq 2u - 3$, and $u \geq 10$ since $q \geq 17$. Increasing u by one adds a single odd integer to $[1, 2u-3]$, so $\pi(q) \leq \pi(2u-3) \leq \pi(17) + (u-10) = u - 3$. Substituting gives the claim. □

7.4. **Larger support primes beyond $n = 1023$.**

Lemma 7.6 (Reciprocal sums)**.** *We have*

$$\sum_{7\le p\le 53}\frac{1}{p}<\frac{259}{400},\qquad \sum_{11\le p\le 149}\frac{1}{p}<\frac{229}{320},\qquad \sum_{11\le p\le 271}\frac{1}{p}<\frac{33}{40}.$$

Proof. Since $\sqrt{271}<17$, an integer in $[17,271]$ is prime exactly when it is coprime to $2\cdot3\cdot5\cdot7\cdot11\cdot13$. So the primes in $[17,271]$ are the integers there that are coprime to 30 and not divisible by 7, 11 or 13. Together with 11 and 13, they are the primes in $[11,271]$. Let B_j be the sum of $1/p$ over these primes in $[30j,30j+30)$. Then:

j	denominators in B_j	$4000B_j <$
0	11, 13, 17, 19, 23, 29	1429
1	31, 37, 41, 43, 47, 53, 59	657
2	61, 67, 71, 73, 79, 83, 89	381
3	97, 101, 103, 107, 109, 113	230
4	127, 131, 137, 139, 149	147
5	151, 157, 163, 167, 173, 179	146
6	181, 191, 193, 197, 199	105
7	211, 223, 227, 229, 233, 239	106
8	241, 251, 257, 263, 269	79
9	271	15

The first sum is less than $\frac{1}{7}+\frac{1429+657}{4000}-\frac{1}{59}=\frac{259}{400}-\frac{19}{206500}$. The second and third are less than $\frac{2844}{4000}=\frac{711}{1000}<\frac{229}{320}$ and $\frac{3295}{4000}=\frac{659}{800}<\frac{33}{40}$. □

Lemma 7.7 (Support primes in $[17,53]$)**.** *If $17\le q\le 53$ and $q<2(u-1)$, then $\mathrm{wt}(F_nD)>n$ for every $n\ge 1024$.*

Proof. By (2.6) (in the equivalent form $4\Phi_q(n)+(u-4)I_q(n)+2\Phi_q(n/3)+2\Phi_q(n/5)$) and (7.1),

$$\mathrm{wt}(F_nD)\ge \alpha_q\Big(\frac{u}{2}+\frac{46}{15}\Big)n-\frac{22u+24}{15}\,\nu_q. \tag{7.2}$$

Use the five bands below. In each band, $u\ge u_*$, $\nu_q\le\nu_*$ and $\alpha_q>a_*$. The last inequality follows from the first sum in Lemma 7.6, after removing the primes beyond the band.

$q\in$	$[17,23]$	$[29,37]$	$[41,43]$	$\{47\}$	$\{53\}$
u_*	10	16	22	25	28
a_*	7/50	7/60	13/125	99/1000	47/500
ν_*	7	10	12	13	14

At $n=1024$ and $u=u_*$, the right side of (7.2), with a_* and ν_* in place of α_q and ν_q, exceeds n by

$$\tfrac{6964}{375},\quad \tfrac{10672}{225},\quad \tfrac{126832}{1875},\quad \tfrac{106162}{1875},\quad \tfrac{40192}{1875}.$$

At $n=1024$ its slopes in u are $\frac{4606}{75}$, $\frac{676}{15}$, $\frac{4456}{125}$, $\frac{11858}{375}$, $\frac{10348}{375}$. Its slope in n, minus one, is positive at u_* and increases with u. So the bound exceeds n for all $n\ge1024$ and $u\ge u_*$. □

Lemma 7.8 (Support primes in $[59,271]$)**.** *If $59\le q\le 271$ and $q<2(u-1)$, then $\mathrm{wt}(F_nD)>n$ for every $n\ge 32768$.*

Proof. Sieve first by $T_0=\{2,3,5,7\}$. This has density $\frac{8}{35}$ and discrepancy at most $\frac{28}{15}$ by Lemma 7.1. With

$$\alpha=\tfrac{8}{35}\Big(1-\sum_{11\le p\le q}\tfrac{1}{p}\Big),\qquad \bar\delta=\tfrac{28}{15}(\pi(q)-3),$$

Lemma 7.2 and (2.5) give

$$\mathrm{wt}(F_nD)\ge 4(\alpha n-\bar\delta)+(u-4)(\tfrac12\alpha n-2\bar\delta)=\tfrac{u+4}{2}\alpha n-2(u-2)\bar\delta.$$

From $\pi(q)\le 2+\Phi_5(q)<3+\frac{4q}{15}$ we get $\bar\delta<\frac{112q}{225}$.

- If $59 \le q \le 149$, then Lemma 7.6 gives $\alpha > \frac{13}{200}$, and $\bar{\delta} < 75$, and the hypothesis gives $u \ge 31$. At $n = 32768$ and $u = 31$ the bound minus n is $\frac{11}{80} \cdot 32768 - 2 \cdot 29 \cdot 75 = \frac{778}{5} > 0$.
- If $151 \le q \le 271$, then $\alpha > \frac{1}{25}$, $\bar{\delta} < 135$ and $u \ge 77$. The corresponding margin is $\frac{31}{50} \cdot 32768 - 2 \cdot 75 \cdot 135 = \frac{1654}{25} > 0$.

In both cases the slopes in n and in u are positive at the corner and remain so; the u-slopes are $\frac{13}{400}n - 150$ and $\frac{n}{50} - 270$. □

Lemma 7.9 (Odd-integer telescoping)**.** *For every odd prime q, $\varrho(q) > 1/(2\sqrt{q})$.*

Proof. Including the odd composite factors decreases the product. For odd $m \ge 3$ we have $(1 - 1/m)^2 > (m-2)/m$, and the product of $(m-2)/m$ over odd $m \in [3, q]$ telescopes to $1/q$. Hence $\varrho(q) \ge \frac{1}{2} \prod_{3 \le m \le q,\ m \text{ odd}} (1 - 1/m) > \frac{1}{2\sqrt{q}}$. □

By Mertens' theorem [11], $\varrho(q) \sim e^{-\gamma}/\log q$. The elementary bound of Lemma 7.9 is weaker but sufficient.

Lemma 7.10 (Large support primes)**.** *Suppose $n \ge 32768$, $u < \frac{20}{7} \log n$ and $277 \le q < 2(u-1)$. Then $\mathrm{wt}(F_n D) > \frac{5}{4}n$.*

Proof. Put $L = \log n$. Then $q < \frac{40}{7}L$ and $L > \frac{7}{40}(q+2) \ge \frac{1953}{40} > 48$. The 11-rough integers up to q include 1, every prime in $(11, q]$, and the composites 169 and 221. By Lemma 7.1 (density $\frac{16}{77}$, discrepancy $\frac{56}{15}$),

$$\pi(q) \le \Phi_{11}(q) + 2 \le \tfrac{16}{77}q + \tfrac{86}{15} < \tfrac{16}{77}q + 6.$$

Since $\log 2 < \frac{7}{10}$, this gives

$$\delta_q < \tfrac{112}{15} \exp(\tfrac{16}{77} q \log 2) < \tfrac{112}{15} e^{64L/77},$$

and hence

$$\frac{2u\delta_q}{n} < \frac{128}{3} L e^{-13L/77} < 2048\, e^{-624/77} < \frac{2048}{2800} < \frac{3}{4}.$$

Here $Le^{-13L/77}$ decreases for $L \ge 48$, $\frac{624}{77} > 8$, and $e^8 = e^6 e^2 > 20^2 \cdot 7 = 2800$. The last bound follows from the partial sums $e^3 > \sum_{j \le 8} 3^j/j! = \frac{89641}{4480} > 20$ and $e^2 > \sum_{j \le 4} 2^j/j! = 7$. On the other hand, $u > q/2$ and Lemma 7.9 give $\frac{1}{2}u\varrho(q) > \frac{\sqrt{q}}{8} > 2$. By (2.5), $\mathrm{wt}(F_n D) \ge u I_q(n) \ge u(\frac{1}{2}\varrho(q)n - 2\delta_q) > 2n - \frac{3}{4}n$. □

7.5. **Proof of Theorem 1.2.** Let $n \ge 67$ and let D be as in the theorem. If $q \le 13$, apply Proposition 7.4. So assume $q \ge 17$, which forces $u \ge 10$. Every prime in $(n/2, n]$ is q-rough, so the index-one components give $\mathrm{wt}(F_n D) \ge u\, r(n)$, and we may assume $u < n/r(n)$.

- For $67 \le n \le 1023$, Proposition 5.1(a) and Lemma 7.5 give $\mathrm{wt}(F_n D) \ge \beta(n) \ge n$.
- For $1024 \le n \le 32768$, Proposition 5.1(b) gives $u < 30$. Hence $q < 2(u-1) \le 56$, so $q \le 53$, and Lemma 7.7 applies.
- For $n \ge 32768$, Proposition 5.1(c) gives $u < \frac{20}{7} \log n$. Lemmas 7.7, 7.8 and 7.10 cover $q \le 53$, $59 \le q \le 271$ and $q \ge 277$, respectively. The only odd integers in the gaps, $55, 57, 273, 275$, are composite.

This completes the proof over every field. □

8. The intervals $n \le 66$

All polynomials in this section are over $\mathbb{F}_2$. We apply the descent of Section 3 to $F = F_n = C_n + Z_n$ with the target $N = n = \mathrm{wt}(C_n) + r(n)$. A counterexample has weight at most

$$m^*(n) = \begin{cases} n - 1, & n \text{ odd}, \\ n - 2, & n \text{ even}, \end{cases}$$

by Lemma 2.3. In the setting of Proposition 3.3, write $m = \mathrm{wt}(G) \le m^*(n)$, $u = \mathrm{wt}(D)$ and $r = r(n)$.

The proof uses only the cases $n \le 10$ as inputs to other cases. Within the range $11 \le n \le 66$, no case is used to prove another.

8.1. **Three further consequences of the descent.**

Lemma 8.1 (Dispersed capacity)**.** *Suppose that one full parity class of C has s terms and every other class has at most $t \le s$ terms. If the fresh-even multiplier D of Proposition 3.3 occupies k full parity classes, then every class of CD has at most $us - (k-1)(s-t)$ terms.*

Proof. A class $(CD)_{[\epsilon]}$ is the sum of the products $C_{[\epsilon-\epsilon'']}D_{[\epsilon'']}$ over the k classes ϵ'' occupied by D, of sizes $u_1, \dots, u_k \ge 1$ with $\sum u_i = u$. At most one of these products uses the largest class of C, say the one with D-class size u_j. So the class has at most $su_j + t(u - u_j)$ terms. This is nondecreasing in u_j, and $u_j \le u - (k-1)$, so it is at most $s(u-k+1) + t(k-1) = us - (k-1)(s-t)$. □

For F_n we may take $s = \lfloor\sqrt{n}\rfloor$ and $t = \lfloor\sqrt{n/2}\rfloor$, by (2.3).

Lemma 8.2 (Unit classes)**.** *Let $Q_1(n)$ be the number of squarefree integers in $(n/4, n]$. In the setting above, if D occupies a single full parity class, then $m \ge u\,Q_1(n)$. Moreover $Q_1(n) > n/4$ for every $n \ge 1$, so if $u \ge 4$ the multiplier D of Proposition 3.3 occupies at least two classes.*

Proof. For squarefree $a \in (n/4, n]$, the class of F_n indexed by a is the single monomial $X(a)$, by (2.3). If D occupies one class, then multiplication by D maps classes injectively to classes, so $X(a)D$ is a full class of F_nD with u terms. If a is not a fresh prime, this is a class of CD of weight $u \le ru$, so it is protected (Corollary 3.5(i)) and appears unchanged in G. If $a = p$ is fresh, the class is x_pD, which is part of $ZD \subseteq G$. These $Q_1(n)$ classes are distinct, so $m \ge uQ_1(n)$.

For the count, let $Q(n)$ be the number of squarefree integers up to n. Every non-squarefree integer is divisible by p^2 for some prime p, so

$$n - Q(n) \le n \sum_p \frac{1}{p^2} < n\Big(\frac{1}{4} + \sum_{k\ge1} \frac{1}{(2k+1)^2}\Big) \le \frac{n}{2},$$

using $\frac{1}{(2k+1)^2} < \frac{1}{4}(\frac{1}{k} - \frac{1}{k+1})$. Thus $Q_1(n) = Q(n) - Q(\lfloor n/4\rfloor) > \frac{n}{2} - \frac{n}{4}$. If $u \ge 4$ and D occupied one class, then $m > un/4 \ge n$, which is impossible. □

Lemma 8.3 (The core cannot vanish cheaply)**.** *Let $C, P_1, \dots, P_r \in \mathbb{F}_2[\mathbf{x}]$ with $C \ne 0$, and put $F = C + \sum_i z_iP_i$. Let $G = FD + F^2E$, where $D \ne 0$ and F^2E are fresh-even. If $\mathrm{wt}(D) < \mathrm{dist}(C)$, then*

$$\mathrm{wt}(G) \ge \mathrm{dist}(C) + \sum_{i=1}^{r} \mathrm{wt}(P_iD).$$

In particular, in the setting of Propositions 3.3 and 3.4, $u < \mathrm{dist}(C)$ implies $m \ge ru + \mathrm{dist}(C)$.

Proof. Since F^2E is fresh-even, so is E. The fresh-odd part of G is $\sum_i z_iP_iD$, with disjoint summands. The fresh-even part is $CD + (C^2 + \sum_i w_iP_i^2)E$, where $w_i = z_i^2$. Regard D and E as polynomials in $w_1, \dots, w_r$ with coefficients in $\mathbb{F}_2[\mathbf{x}]$. Let j be the least total w-degree occurring in D or E.

The degree-j part of the fresh-even part of G is $C(D_j + CE_j)$, and it is nonzero. Indeed, if $D_j = CE_j$, then either $E_j \ne 0$ and some coefficient of D_j is a nonzero multiple of C with fewer than $\mathrm{dist}(C)$ terms, or $E_j = 0 = D_j$, contrary to the choice of j. So some w-coefficient of the fresh-even part is a nonzero multiple of C, contributing at least $\mathrm{dist}(C)$ terms.

For the final assertion, take $P_i = 1$ and $F^2E = R$. □

8.2. **Cores.** In the rest of this section write $x = x_2$ and $y = x_3$, and put $c_k = 1 + x + \cdots + x^k$. Note that $c_3 = c_1^3$ and $c_1c_4 = 1 + x^5$, and that

$$F_2 = c_1, \qquad F_3 = c_1 + y, \qquad F_4 = c_2 + y.$$

Lemma 8.4 (A two-row obstruction)**.** *Let $P = \sum_{i=0}^{d} a_i y^i$ with coefficients $a_i \in \mathbb{F}_2[\mathbf{x}]$, $d \geq 1$ and $a_0 a_d \neq 0$. Suppose some nonzero multiple of P has only two nonzero y-coefficients. Then*

$$a_i^{d-j} a_d^{j-i} = a_j^{d-i} \qquad \text{whenever } 0 \leq i < j < d \text{ and } a_i a_j \neq 0.$$

Hence, if one such identity fails, $\operatorname{dist}(P) \geq \operatorname{dist}(a_0) + \operatorname{dist}(a_d) + 1$. *For a quadratic with $a_1 \neq 0$, it suffices that $a_0 a_2 \neq a_1^2$.*

Proof. Let Ω be an algebraic closure of $\mathbb{F}_2(\mathbf{x})$. After removing a power of y, the multiple is $b_0 + b_1 y^h$ with $b_0 b_1 \neq 0$. Since $y \nmid P$, the polynomial P divides $b_0 + b_1 y^h$ in $\mathbb{F}_2(\mathbf{x})[y]$. The roots of the latter in Ω are $\rho\zeta$, where ρ is one root and ζ runs over h-th roots of unity. So the roots of P, with multiplicity, are $\rho\zeta_1, \ldots, \rho\zeta_d$ with each ζ_i algebraic over $\mathbb{F}_2$.

Hence $a_i/a_d = \rho^{d-i}\gamma_i$, where γ_i is an elementary symmetric function of the ζ's and is algebraic over $\mathbb{F}_2$. The quotient $a_i^{d-j} a_d^{j-i} / a_j^{d-i} = \gamma_i^{d-j} / \gamma_j^{d-i}$ lies in $\mathbb{F}_2(\mathbf{x})$ and is algebraic over $\mathbb{F}_2$. Since $\mathbb{F}_2$ is algebraically closed in the purely transcendental extension $\mathbb{F}_2(\mathbf{x})$, the quotient lies in $\mathbb{F}_2^{\times} = \{1\}$.

Finally, a nonzero multiple of P has y-width at least $d \geq 1$, so it has at least two nonzero rows. If an identity fails it has at least three rows: the lowest is a multiple of a_0, the highest a multiple of a_d, and a third contributes at least one term. □

Define the cores

$$f_9 = c_3 + c_1 y + y^2, \qquad f_{12} = c_3 + c_2 y + y^2, \qquad f_{16} = c_4 + c_2 y + y^2,$$
$$f_{18} = c_4 + c_2 y + c_1 y^2, \qquad f_{24} = c_4 + c_3 y + c_1 y^2, \qquad f_{25} = f_{24} + x_5 F_4 + x_5^2.$$

Here f_k encodes the 3-smooth integers up to k for $k \leq 24$, and f_{25} the 5-smooth integers up to 25. Writing $f_{16}(x, x_5)$ for f_{16} with y replaced by x_5, one checks that

$$f_{25} = f_{16}(x, x_5) + (c_3 + x_5)\, y + c_1 y^2. \tag{8.1}$$

Lemma 8.5 (Core bounds)**.** $\operatorname{dist}(f_9) \geq 5$, $\operatorname{dist}(f_{12}) \geq 4$, $\operatorname{dist}(f_{16}) \geq 4$, $\operatorname{dist}(f_{18}) \geq 6$, $\operatorname{dist}(f_{24}) \geq 5$ *and* $\operatorname{dist}(f_{25}) \geq 8$.

Proof. For all but f_9 we use Lemma 2.2, Lemma 8.4 and Lemma 2.3. The endpoint bounds in y are $2+1$, $2+1$, $2+2$, $2+2$ for $f_{12}, f_{16}, f_{18}, f_{24}$, using $\operatorname{dist}(c_k) \geq 2$. The quadratic identities fail:

$$c_3 \neq c_2^2, \qquad c_4 \neq c_2^2, \qquad c_1 c_4 = 1 + x^5 \neq c_2^2, \qquad c_1 c_4 \neq c_3^2.$$

So the two-row obstruction adds one. Moreover f_{18} has even weight 10, so its bound 5 rounds up to 6.

For f_{25} use (8.1). The endpoint bound is $\operatorname{dist}(f_{16}) + \operatorname{dist}(c_1) \geq 4 + 2$. The identity fails because $c_1 f_{16}(x, x_5)$ has the nonzero linear x_5-coefficient $c_1 c_2$, whereas $(c_3 + x_5)^2$ has none. This gives 7, and the even weight 16 of f_{25} rounds it to 8.

For f_9 we use a parametrisation. By Lemma 2.1(b) it suffices to consider multipliers in $\mathbb{F}_2[x, y]$. Over $\mathbb{F}_2$, substitute

$$x = v^2 + v + 1, \qquad y = v^2(v+1),$$

so that $c_1 = v(v+1)$ and $f_9 = v^3(v+1)^2((v+1) + 1 + v) = 0$. At $v = 0$ and $v = 1$ the value of x is 1, while y vanishes to orders 2 and 1, respectively. For $k > 0$ the polynomial $1 + x^k$ vanishes to order $2^{v_2(k)}$ at both points, since $1 + x = v(v+1)$ has simple zeros there.

Suppose a multiple G of f_9 had at most four terms, normalised to have a term free of y. Evaluating at $v = 0$ shows that its y-free part has a positive even number of terms. That part cannot be all of G, since $x(v)$ is transcendental. So it is a binomial $x^i(1 + x^k)$, whose orders at $v = 0$ and $v = 1$ are equal. At most two terms involve y.

- If a unique term has the least positive y-exponent e, these terms have orders $(2e, e)$ at the two points.
- If two terms share that exponent, they have orders $(2e + \tau, e + \tau)$ with $\tau = 2^{v_2(k')}$ for some k'.

In neither case are the two orders equal, so the y-part cannot cancel the y-free part at both points. Hence $\operatorname{dist}(f_9) \geq 5$. □

The bound $\operatorname{dist}(f_9) \geq 5$ is sharp: $f_9(1+x+y) = 1 + xy + x^3y + y^3 + x^4$, that is, $\{1,2,3\} * \{1,2,3,4,6,8,9\} = \{1,6,16,24,27\}$.

8.3. **The cases $n \leq 10$.** For $n \leq 3$ the endpoint bound gives $\operatorname{dist}(F_n) = n$. For $n = 4$, it gives 3, and Lemma 2.3 rounds this up to 4. For $5 \leq n \leq 10$:

n	$r(n)$	C_n	$\kappa \leq \operatorname{dist}(C_n)$
5	2	c_2	2
6	1	$c_2 + yF_2$	4
7	2	$c_2 + yF_2$	4
8	2	$c_3 + yF_2$	4
9	2	f_9	5
10	1	$f_9 + x_5F_2$	7

The bounds come from Lemmas 2.2 and 8.5 and $\operatorname{dist}(F_2) = 2$. Suppose $\operatorname{dist}(F_n) < n$, and take $G = G_0 + ZD$ as in Proposition 3.3 with $N = n$. Then $ru < m \leq m^*(n)$ and $u \leq m/2$ give $u < \kappa$ in every row. So Lemma 8.3 gives $m \geq ru + \kappa$. At $n = 6$ this exceeds $m^*(6) = 4$. In the other rows it forces $u = 1$, and then $\operatorname{wt}(F_nD) = n$. Every class of C_nD has at most $M = \sigma(C_n) = \lfloor \sqrt{n} \rfloor$ terms, and $n \geq 2M$, so Corollary 3.5(iii) excludes the counterexample. Hence $\operatorname{dist}(F_n) = n$ for all $n \leq 10$. In particular $\operatorname{dist}(F_2) = 2$, $\operatorname{dist}(F_3) = 3$ and $\operatorname{dist}(F_4) = 4$.

8.4. **Few multipliers and few primes.** The results of this subsection hold for every $n \geq 1$ and every multiplier, including multipliers involving fresh-prime variables.

One and two multipliers [10]. One multiplier gives exactly n terms. For distinct a, b, after dividing by $\gcd(a,b)$ the larger quotient is at least 2. So $|a[n] \cap b[n]| \leq \lfloor n/2 \rfloor$ and $|\{a,b\} * [n]| \geq 2n - 2\lfloor n/2 \rfloor \geq n$.

Lemma 8.6 (Two primes)**.** *If the normalised multiplier involves at most two primes, then* $\operatorname{wt}(F_nD) \geq n$.

Proof. We use the decomposition (2.4) with T the set of these primes.

One prime p, at least two terms. The component of b is D itself when $\lfloor n/b \rfloor < p$. Otherwise it has at least two terms by Lemma 2.1(c). There are $n - \lfloor n/p \rfloor \geq \lceil n/2 \rceil$ components, so $\operatorname{wt}(F_nD) \geq 2\lceil n/2 \rceil \geq n$.

Two primes $p < q$, at least three terms. Fewer terms are covered by the one- and two-multiplier bounds above. Every component has at least three terms:

- index $j < p$: the component is D;
- $p \leq j < q$: two x_q-slices, each with at least two terms by Lemma 2.2;
- $j \geq q$: three endpoints, namely 1, x_p and x_q.

So $\operatorname{wt}(F_nD) \geq 3\Phi_{\{p,q\}}(n)$, and it suffices to bound the number $\Phi_{\{p,q\}}(n)$ of integers up to n divisible by neither p nor q.

- If $p \geq 3$, then $\Phi_{\{p,q\}}(n) \geq n(1 - \frac{1}{3} - \frac{1}{5}) > n/3$.
- If $p = 2$ and $q \geq 5$, then $\Phi_{\{2,q\}}(n) = \lceil n/2 \rceil - \lceil \lfloor n/q \rfloor / 2 \rceil$ is nondecreasing in q. For $q = 5$, the residues $1, 3, 7, 9$ modulo 10 give $3\Phi \geq n$: at the ends $2, 6, 8, 10$ of the constant stretches in the first period the values of 3Φ are $3, 6, 9, 12$, and each period adds two to the surplus.
- If $p = 2$ and $q = 3$, then $3\Phi_{\{2,3\}}(n) \geq n$ fails only for $n \equiv 4 \pmod 6$, and then by one. For $n = 6k + 4 \geq 10$ an integer coprime to 6 lies in $(n/3, n/2]$: take 5 or 7 when $k = 1, 2$; when $k \geq 3$, the integers $2k + 2, \ldots, 3k + 2$ include four consecutive ones. Its component has index two and hence at least four terms, which supplies the missing term. The case $n = 4$ is $\operatorname{dist}(F_4) = 4$.

□

Corollary 8.7 (Three multipliers [10])**.** *Every multiplier with at most three terms gives at least n terms.*

Proof. Take a counterexample with at most three terms, minimal under deletion and normalised, with u terms. The one- and two-multiplier bounds give $u = 3$. Lemma 4.1 gives $q < 4$ for every prime involved, so only 2 and 3 occur. This contradicts Lemma 8.6. □

Lemma 8.8. *For distinct primes $p < q < \ell$ and every $n \geq 11$, the number $\Phi_{\{p,q,\ell\}}(n)$ of integers up to n coprime to $pq\ell$ satisfies $4\Phi_{\{p,q,\ell\}}(n) \geq n$.*

Proof.
- If $p \geq 3$, the union bound gives at least $n(1 - \frac{1}{3} - \frac{1}{5} - \frac{1}{7}) = \frac{34}{105}n > \frac{n}{4}$.
- If $p = 2$, $q \geq 5$ and $\ell \geq 7$, subtracting the odd multiples of q and ℓ from the odd integers gives at least $(\frac{1}{2} - \frac{1}{10} - \frac{1}{14})n - 1 = \frac{23}{70}n - 1 \geq \frac{n}{4}$ for $n \geq 13$. For $n = 11, 12$, each of q and ℓ removes at most one of the six odd integers.
- If $p = 2$ and $q = 3$, then $\Phi_{\{2,3,\ell\}}(n) = \Phi_{\{2,3\}}(n) - \Phi_{\{2,3\}}(n/\ell) \geq \Phi_5(n)$. This count increases by 8 per period of 30. At the ends $6, 10, 12, 16, 18, 22, 28, 30$ of its constant stretches it takes the values $1, \ldots, 8$. So $4\Phi_5(n) - n < 0$ in the first period only at $n = 5, 6, 9, 10$, and each further period adds two.

□

Theorem 8.9 (Three primes)**.** *If the normalised multiplier involves at most three primes, then $\mathrm{wt}(F_n D) \geq n$ for every $n \geq 1$.*

Proof. By the cases $n \leq 10$, Lemma 8.6 and Corollary 8.7, we may assume $n \geq 11$, $u \geq 4$, and that exactly three primes $p < q < \ell$ occur. Every component has at least four terms:

- index $j < p$: the component is D;
- $p \leq j < q$: at least two joint (x_q, x_ℓ)-slices, each with two endpoints;
- $q \leq j < \ell$: at least two x_ℓ-slices, each with three endpoints;
- $j \geq \ell$: four endpoints.

Now apply Lemma 8.8. □

Corollary 8.10 (Four multipliers)**.** *Every multiplier with at most four terms gives at least n terms, for every n.*

Proof. A minimal normalised counterexample would have $u = 4$ by Corollary 8.7. By Lemma 4.1 it would involve only primes $q < 6$, contradicting Theorem 8.9. □

8.5. **The cases $11 \leq n \leq 26$.** Write the cores as a 3- or 5-smooth part plus linear terms in larger primes. The bound κ adds the core bound of Lemma 8.5 to $\mathrm{dist}(F_j) = j$ for each linear coefficient F_j, by Lemma 2.2. Put $U = \lfloor (m^*(n) - \kappa)/r \rfloor$.

n	r	C_n	κ	U
11	2	$f_9 + x_5F_2$	7	1
12	2	$f_{12} + x_5F_2$	6	2
13	3	$f_{12} + x_5F_2$	6	2
14	2	$f_{12} + x_5F_2 + x_7F_2$	8	2
15	2	$f_{12} + x_5F_3 + x_7F_2$	9	2
16	2	$f_{16} + x_5F_3 + x_7F_2$	9	2
17	3	$f_{16} + x_5F_3 + x_7F_2$	9	2
18	3	$f_{18} + x_5F_3 + x_7F_2$	11	1
19	4	$f_{18} + x_5F_3 + x_7F_2$	11	1
20	4	$f_{18} + x_5F_4 + x_7F_2$	12	1
21	4	$f_{18} + x_5F_4 + x_7F_3$	13	1
22	3	$f_{18} + x_5F_4 + x_7F_3 + x_{11}F_2$	15	1
23	4	$f_{18} + x_5F_4 + x_7F_3 + x_{11}F_2$	15	1
24	4	$f_{24} + x_5F_4 + x_7F_3 + x_{11}F_2$	14	2
25	4	$f_{25} + x_7F_3 + x_{11}F_2$	13	2
26	3	$f_{25} + x_7F_3 + x_{11}F_2 + x_{13}F_2$	15	3

In every row, $ru < m \le m^*(n)$ and $u \le m/2$ give $u < \kappa$. Then Lemma 8.3 gives $m \ge ru + \kappa$, hence $u \le U$. Every class of C_nD has at most $M = u\lfloor\sqrt{n}\rfloor \le U\lfloor\sqrt{n}\rfloor$ terms, and $\mathrm{wt}(F_nD) \ge n$ by Corollary 8.7. Every row except $n = 26$ satisfies $n \ge 2U\lfloor\sqrt{n}\rfloor$, and at $n = 26$ the inequality $n \ge 2u\lfloor\sqrt{n}\rfloor$ holds for $u \le 2$. So Corollary 3.5(iii) excludes everything except $n = 26$ with $u = 3$, which is treated together with $27 \le n \le 29$ below.

8.6. **The cases $27 \le n \le 66$.** Let B_n be the sum of $X(a)$ over the $\sqrt{n}$-smooth $a \le n$. The primes p with $\sqrt{n} < p \le n/2$ occur in C_n only linearly:

$$C_n = B_n + \sum_{\sqrt{n}<p\le n/2} x_p F_{\lfloor n/p\rfloor}.$$

Slicing in these variables (Lemma 2.2) gives

$$\mathrm{dist}(C_n) \ge \kappa_B(n) + \sum_{\sqrt{n}<p\le n/2} \Big\lfloor \frac{n}{p} \Big\rfloor. \tag{8.2}$$

Here we use $\mathrm{dist}(F_j) = j$ for $j = \lfloor n/p\rfloor < \sqrt{66} < 9$, which is covered by the cases $n \le 10$. The bounds $\kappa_B(n) \le \mathrm{dist}(B_n)$ are:

n	27–29	30–31	32–35	36–39	40–44	45–48	49–66
$\kappa_B(n)$	7	8	7	8	7	6	8

Proof of the values of κ_B. Regard B_n as a polynomial in $x = x_2$, and apply Lemma 8.4 to its top three coefficients, using $i = d - 2$ and $j = d - 1$. The top coefficient is always 1. The bottom coefficient, which encodes the odd $\sqrt{n}$-smooth integers up to n, is as follows.

- For $27 \le n \le 44$ it is $f_9(y, x_5)$, of distance at least 5.
- For $45 \le n \le 48$ it is $f_{12}(y, x_5)$, of distance at least 4.
- For $49 \le n \le 66$ it is $g = f_{12}(y, x_5) + hx_7 + x_7^2$, with $h = 1 + y + x_5$ for $n \le 62$ and $h = 1 + y + y^2 + x_5$ for $n \ge 63$. Since $f_{12}(y, x_5)$ has a term of odd exponent, it is not h^2, so $\mathrm{dist}(g) \ge 4 + 1 + 1 = 6$ by Lemma 8.4.

The degree $d = \deg_x B_n$ and the coefficients (a_{d-2}, a_{d-1}, a_d) are:

n	d	a_{d-2}	a_{d-1}	a_d
27–31	4	$1 + y + x_5$	$1 + y$	1
32–39	5	$1 + y$	1	1
40–47	5	$1 + y + x_5$	1	1
48–55	5	$1 + y + x_5$	$1 + y$	1
56–63	5	$1 + y + x_5 + x_7$	$1 + y$	1
64–66	6	$1 + y$	1	1

In each row $a_{d-2}a_d \ne a_{d-1}^2$, so Lemma 8.4 adds one to the endpoint bound. This gives $5 + 1 + 1 = 7$ for $27 \le n \le 44$, $4 + 1 + 1 = 6$ for $45 \le n \le 48$, and $6 + 1 + 1 = 8$ for $49 \le n \le 66$. The weights of B_n are $17, 18, 19, 20, 21$ on the ranges 27–29, 30–31, 32–35, 36–39, 40–44. On the two even-weight ranges, Lemma 2.3 raises 7 to 8. □

Regular blocks. For a block $[L, H]$ of consecutive n put

$$r_0 = \pi(L) - \pi(H/2), \qquad \kappa_0 = \min_{L\le n\le H} \kappa_B(n) + \sum_{\sqrt{H}<p\le L/2} \Big\lfloor \frac{L}{p} \Big\rfloor, \qquad U = \Big\lfloor \frac{m^*(H) - \kappa_0}{r_0} \Big\rfloor.$$

Then $r(n) \ge r_0$, $\mathrm{dist}(C_n) \ge \kappa_0$ and $m \le m^*(H)$ for $n \in [L, H]$. In every block below, $\lfloor (m^*(H) - 1)/r_0\rfloor < \kappa_0$, so $ru < m$ gives $u < \kappa_0$, and Lemma 8.3 gives $u \le U \le 4$. Put $s = \lfloor\sqrt{H}\rfloor$, and let $M_0 = Us$ if $U \le 3$, and $M_0 = 3s + \lfloor\sqrt{H/2}\rfloor$ if $U = 4$.

By Lemmas 8.1 and 8.2, M_0 bounds every class of C_nD. Indeed $u\lfloor\sqrt{n}\rfloor \le us \le Us = M_0$ if $U \le 3$, and $us \le 3s \le M_0$ if $U = 4$ and $u \le 3$. If $u = 4$, then D occupies at least two classes, so Lemma 8.1 gives the bound $3\lfloor\sqrt{n}\rfloor + \lfloor\sqrt{n/2}\rfloor \le M_0$. Every block has $L \ge 2M_0$. Moreover $\mathrm{wt}(F_nD) \ge n$ by Corollary 8.10. So Corollary 3.5(iii) excludes all these blocks.

$[L,H]$	r_0	κ_0	U	s	$[L,H]$	r_0	κ_0	U	s
30–32	4	15	3	5	41–42	5	22	3	6
33	5	16	3	5	43–45	6	22	3	6
34	4	18	3	5	46	5	25	3	6
35	4	19	3	5	47–48	6	25	3	6
36	4	20	3	6	53–57	7	23	4	7
37	5	20	3	6	58	6	27	4	7
38	4	22	3	6	59–61	7	27	4	7
39	4	23	3	6	62–63	7	29	4	7
					64	7	29	4	8
					65–66	7	30	4	8

The primes used are those up to 61, all detected by trial division by $2, 3, 5, 7$. The remaining values of n are covered by the same preliminary inequalities, with the following data. Here the comparison with $M = us$ already excludes $u \le 2$ at $n = 27$–29, and $u \le 3$ at $n = 40$ and $n = 49$–52.

n	r_0	κ_0	U
27	3	14	4
28	3	15	3
29	4	15	3
40	4	22	4
49–50	6	21	4
51–52	6	22	4

Three terms at $26 \le n \le 29$. The ten squarefree integers $10, 11, 13, 14, 15, 17, 19, 21, 22, 23$ lie in $(n/4, n]$, so $Q_1(n) \ge 10$. If $u = 3$ and D occupied one class, Lemma 8.2 would give $m \ge 30 > n$. So D occupies at least two classes, and Lemma 8.1 gives

$$M = 2\lfloor\sqrt{n}\rfloor + \lfloor\sqrt{n/2}\rfloor = 13.$$

Since $n \ge 26 = 2M$ and $\mathrm{wt}(F_nD) \ge n$ by Corollary 8.7, Corollary 3.5(iii) applies.

Four terms at $n = 40$ *and* $49 \le n \le 52$.

Lemma 8.11. *Every four-term multiplier* D *satisfies* $\mathrm{wt}(F_{40}D) \ge 44$*, and* $\mathrm{wt}(F_nD) \ge 56$ *for* $49 \le n \le 52$.

Proof. Normalise D and let q be its largest prime.

$q \ge 7$. Split D at the first positive x_q-exponent, as in the proof of Lemma 4.1. Both pieces have at most three terms, so each gives at least n terms by Corollary 8.7. The intersection has at most $3\lfloor n/7\rfloor$ elements. Hence $\mathrm{wt}(F_nD) \ge 2n - 6\lfloor n/7\rfloor$, which is 50 at $n = 40$ and at least 56 for $49 \le n \le 52$.

$q = 2$. The components give $\mathrm{wt}(F_nD) \ge 2\Phi_2(n) + 2I_2(n) \ge \frac{3}{2}n - 1$.

$q = 3$. Every component has at least four terms:

- index one: D itself;
- index two: two x_3-slices with two endpoints each;
- larger indices: three endpoints, rounded up to four by Lemma 2.3, since D has even weight.

Since $\Phi_3(40) = 13$ and $\Phi_3(49) = 17$, this suffices.

$q = 5$. Every component has at least four terms: two x_5-slices before 5 is active, and four endpoint directions after. Since $\Phi_5(49) = 14$, this settles $49 \le n \le 52$. At $n = 40$, $\Phi_5(40) = 10$. The $\Phi_5(40/3) - \Phi_5(8) = 2$ components of index 3 or 4 have at least six terms each, from two x_5-slices with three endpoints each. The total is at least 44. □

At $n = 40$ we have $r = 4$, $m \le 38$ and $M = 3 \cdot 6 + 4 = 22$ by Lemmas 8.1 and 8.2. Two residual classes would need $2(n - M) = 36$ terms in G_0. But $\mathrm{wt}(G_0) \le 38 - 16 = 22$, so $a \le 1$. Since $\mathrm{wt}(F_{40}D) \ge 44 = 2M$, Corollary 3.5(iii) applies.

At $49 \le n \le 52$ we have $r = 6$, and $M = 26$ works for all four values. Two residual classes would need $2(n-26)$ terms, but $\mathrm{wt}(G_0) \le m - 24 \le n - 25$. So $a \le 1$, and $\mathrm{wt}(F_n D) \ge 56 \ge \max\{n, 52\}$.

Four terms at $n = 27$. We use a small compatibility bound. If $E \ne 0$ has $k \le 3$ terms, then

$$\mathrm{wt}(F_2 E) + \mathrm{wt}(F_3 E) \ge k + 4. \tag{8.3}$$

For $k = 1$ the two weights are 2 and 3. For $k = 2$ they are at least 2 and 4. For $k = 3$, if $\mathrm{wt}(F_2 E) \ge 4$, the endpoint bound $\mathrm{wt}(F_3 E) \ge 3$ suffices. Otherwise $(1+x)E$ has two terms. These agree at $x = 1$, so up to a monomial factor $E = 1+x+x^2$. Then $F_3 E = 1+x^3+y(1+x+x^2)$ has five terms, and the total is $2 + 5$.

Lemma 8.12. *Every four-term multiplier D satisfies* $\mathrm{wt}(F_{27} D) \ge 36$.

Proof. Normalise D and let q be its largest prime. For $q \ge 7$ the split gives $54 - 6 \cdot 3 = 36$. For $q = 2$ the bound is $2\Phi_2(27) + 2I_2(27) = 42$, and for $q = 3$ it is $4\Phi_3(27) = 36$.

For $q = 5$, the 5-rough $b \le 27$ are $1, 7, 11, 13, 17, 19, 23$, with indices $27, 3, 2, 2, 1, 1, 1$.

- The index-27 component has at least 8 terms, since $\mathrm{dist}(B_{27}) \ge 7$ and D has even weight.
- The three index-one components contribute 12.
- Write $D = \sum_i x_5^i E_i$. At least two E_i are nonzero, each with at most three terms. Since F_2 and F_3 do not involve x_5, summing (8.3) over the slices gives $\mathrm{wt}(F_2 D) + \mathrm{wt}(F_3 D) \ge 4 + 4 \cdot 2 = 12$.
- The second index-two component contributes at least 4, two from each of two slices.

The total is at least $8 + 12 + 12 + 4 = 36$. □

Now let $n = 27$ and $u = 4$. Then $\kappa_0 = 14$ forces $m \ge 3 \cdot 4 + 14 = 26 = m^*(27)$, so $\mathrm{wt}(G_0) = 14$. By Lemma 8.2, D occupies at least two classes. The largest class of C_{27} has 5 terms (the squares) and every other class has at most 3. So Lemma 8.1 gives $M = 3 \cdot 5 + 3 = 18$. Two residual classes would need $2(27 - 18) = 18 > 14$ terms, so $a \le 1$. Since $\mathrm{wt}(F_{27} D) \ge 36 = 2M$ by Lemma 8.12, Corollary 3.5(iii) applies.

This completes the proof of Theorem A for $n \le 66$, and with Section 5, for all n. □

9. Remarks and open problems

9.1. **Sharpness of the hypotheses.**

(a) The hypothesis $r \ge \sigma(C)$ of Theorem 3.6 cannot simply be dropped (Example 3.7). For intervals it is what forces the threshold $n \ge 67$, since $r(66) = 7 < \lfloor \sqrt{66} \rfloor$.

(b) Theorem A uses the structure of $[n]$ in an essential way. Holdridge and Pach [8] observe that $\{1, 2\} * \{1, 2, 4\} = \{1, 8\}$, so the analogue of Theorem A fails for the divisor-closed set $\{1, 2, 4\}$. By contrast, Theorem 1.1 applies to every finite set B with enough fresh primes.

(c) In characteristic two, Theorem A shows that the conclusion of Theorem 1.2 holds even without its hypotheses. The hypotheses serve to make a direct counting proof possible over every field. The condition $q \le n/2$ makes every prime in $(n/2, n]$ q-rough. The condition $q < 2(u-1)$ guarantees enough terms in each index-one component. Over other fields, Question 9.3 has since been answered by a different route [20].

9.2. **The structure of minimal counterexamples.** The directed cut is a special case of a general inequality for products of polynomials. The general inequality below is not needed for Theorem A, but it describes the rigidity that any minimal counterexample, to Theorem A or to related statements, must have.

Lemma 9.1 (Separated blocks)**.** *Let $\mathbb{K}$ be a field, $P \ne 0$ a polynomial and $N \ge 1$. Let $D \ne 0$ have u terms with* $\mathrm{wt}(PD) < N$*, and suppose* $\mathrm{wt}(PE) \ge N$ *for every polynomial $E \ne 0$*

obtained by deleting some, but not all, terms of D. Fix a variable x with $\operatorname{ord}_x P = \operatorname{ord}_x D = 0$. For $g \geq 1$ let $\tau_g(P)$ be the number of terms of P with x-exponent at least g. Partition the occupied x-exponents of D into $k+1 \geq 2$ consecutive blocks, each separated from the next by a gap of at least g, and let v be the number of terms of D in the highest block. Then

$$kN < 2(u - v)\,\tau_g(P).$$

Proof. Let $D_1, \dots, D_{k+1}$ be the parts of D in the blocks, in increasing order. Each is proper, so $\operatorname{wt}(PD_i) \geq N$. If an output monomial occurs in $h \geq 2$ of the products PD_i, at most $h \leq 2(h-1)$ terms are lost there on addition. Charge two to each occurrence except the one in the highest participating block.

A charged occurrence in block i has x-exponent at least the least exponent of a higher block. That is at least g more than every exponent in D_i, so it arises from a term of P with x-exponent at least g. There are at most $\operatorname{wt}(D_i)\tau_g(P)$ such occurrences, and the highest block is never charged. Hence $\operatorname{wt}(PD) \geq (k+1)N - 2(u-v)\tau_g(P)$, and comparison with $\operatorname{wt}(PD) < N$ proves the claim. □

Proposition 9.2 (Factorial localisation)**.** *Let D be a normalised counterexample at n that is minimal under deletion, with u terms, over any field. For a prime q, list the occupied x_q-exponents as $0 = a_0 < a_1 < \cdots < a_s$, and let $k_g = \#\{i : a_i - a_{i-1} \geq g\}$. Then*

$$k_g \leq \Big\lfloor \frac{2u-3}{q^g} \Big\rfloor \qquad (g \geq 1), \qquad \textit{hence} \qquad a_s \leq v_q((2u-3)!).$$

In particular every integer encoded by a monomial of D divides $(2u-3)!$.

Proof. Apply Lemma 9.1 with $P = F_n$, $N = n$, $x = x_q$, splitting at all k_g gaps of size at least g. Here $\tau_g(F_n) = \lfloor n/q^g \rfloor$. If $k_g > 0$, then $k_g n < 2(u-1)n/q^g$, so the integer $q^g k_g$ is at most $2u-3$. Summing over g gives $a_s = \sum_{g \geq 1} k_g \leq \sum_g \lfloor (2u-3)/q^g \rfloor = v_q((2u-3)!)$, by Legendre's formula. □

The slice counts for different primes are not independent. For example, $1 + x_2 x_3$ has two slices in each of x_2 and x_3 but only two terms. So Proposition 9.2 does not by itself bound u in terms of the number of primes involved.

9.3. **Open problems.**

Question 9.3 (Other characteristics; since resolved)**.** Is $\operatorname{dist}_{\mathbb{K}}(F_n) = n$ for every field $\mathbb{K}$? Equivalently, does every nonzero finitely supported $f \colon \mathbb{N} \to \mathbb{K}$ have $|\operatorname{supp}(f * \mathbf{1}_{[n]})| \geq n$, where $*$ is multiplicative convolution?

By Corollary 3.9 the answer is yes in characteristic two. Our descent relies on the Frobenius map $P \mapsto P^2$. The question has since been answered affirmatively, for every field and all nonzero weights, by a descent in widths and exponent residues [20].

Question 9.4 (Equality)**.** Which finite sets A satisfy $|A * [n]| = n$? Examples include $\{a\}$, $\{a, 2a\}$ for even n, and the families of Ke and Meyer [10], which in polynomial form are the identities $F_n \cdot F_n^{2^k-1} = F_n^{2^k}$, $k \geq 1$.

Density (since resolved). Holdridge and Pach [8, Theorem 1.8] proved that the limit $h(A) = \lim_n |A * [n]|/n$ satisfies $h(A) \geq \frac{8}{7}$ unless A is $\{a\}$ or $\{a, 2a\}$, and suggested that $\frac{4}{3}$ is the truth [8, Remark 1.9]. A companion note in the same archive [20] proves $h(A) \geq \frac{4}{3}$ for all other A, by a direct divisibility argument rather than the descent of Section 3.

Question 9.5 (Other sets)**.** For which finite $B \subset \mathbb{N}$ does $\min_{A \neq \emptyset} |A * B| = |B|$ hold? Theorem 1.1 reduces this question to T-free multipliers whenever B has at least $\sigma(B \setminus T)$ fresh primes.

Finally, the counting in Sections 6 and 7 is the longest part of the proof. A proof of Theorem 1.2, or of the weaker statement actually used in Section 5, that avoids explicit prime estimates would shorten the argument considerably.

Appendix A. Logical structure

For the reader's convenience, and as a guide to formal verification, we list the inputs of each main step.

Result	Inputs
Lemmas 2.1 to 2.3, (2.3), (2.4), Lemma 2.4	Unique factorisation and elementary polynomial algebra.
Lemma 3.2, Propositions 3.3 and 3.4, Corollary 3.5	Lemmas 2.1 and 2.2; formal derivatives; Frobenius in characteristic two.
Theorem 3.6, Theorem 1.1, Corollary 3.8	Propositions 3.3 and 3.4, Corollary 3.5(i), (2.3).
Lemma 4.1	Minimality under deletion; Lemma 2.1.
Proposition 5.1	Lemmas 6.1, 6.2, 6.4 and 6.5 and Proposition 6.3.
Theorem 1.2	Lemma 2.4, (2.5), (2.6), Lemmas 7.1 to 7.3 and 7.5 to 7.10 and Proposition 7.4, Proposition 5.1.
Theorem A, $n \geq 67$	Proposition 5.1, Corollary 3.8, Lemma 4.1, Theorem 1.2. No other value of n.
Theorem A, $n \leq 10$	Lemmas 2.2, 2.3 and 8.3 to 8.5, Proposition 3.3, Corollary 3.5(iii).
Lemma 8.6, Corollary 8.7, Theorem 8.9, Corollary 8.10	Theorem A for $n \leq 10$; Lemmas 4.1 and 8.8.
Theorem A, $11 \leq n \leq 66$	Lemmas 8.1 to 8.3, 8.5, 8.11 and 8.12, Corollaries 8.7 and 8.10, Corollary 3.5; Theorem A for $j \leq 8$ (for the coefficients F_j). No case $n' \geq 11$.

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We thank Philippa Holdridge and Péter Pál Pach, whose paper [8] introduced several of the ideas developed here. In particular it introduced the slicing at the primes in $(n/2, n]$, the use of squares in characteristic two, and the rough-component and splitting estimates. The pairing that underlies the use of squares goes back to Ke and Meyer [10]. More broadly, this work rests on the contributions of Pilz [17], Huang, Ke and Pilz [9], Pach and Szabó [14], Ke and Meyer [10], Nagy and Pach [13], and Holdridge and Pach [8], described in the introduction. Any errors are ours.

Subsequent work

The method of this paper turned out not to depend on characteristic two. A subsequent paper [20] proves that for every field K and all nonzero weights $c_1, \ldots, c_n \in K$, every nonzero multiple of $\sum_{a \leq n} c_a X(a)$ has at least n terms. Theorem A is the case $K = \mathbb{F}_2$, $c_a = 1$. That proof replaces the square-root descent with a descent in widths and exponent residues, does not use Theorem A as an input, and has its own formalisation, which shares this one's arithmetic estimates; the two formal statements of Theorem A are proved identical. A companion note in the same archive proves the density bound $h(A) \geq 4/3$ anticipated in [8, Remark 1.9].

How this work was produced

The arguments in this paper, its written proofs, and its formalisation were developed with substantial assistance from AI systems. The written proofs have not been checked line by line by a human mathematician. The Lean 4 formalisation is over $\mathbb{F}_2$: Theorem A and the intermediate results it needs are machine-checked there, using only the standard axioms of Lean's foundations, and where the text and the formal proof differ, the formal proof is authoritative. The statements made here over general fields, or with vector coefficients (Theorem 1.2 as stated, Corollary 3.9, Lemma 9.1 and Proposition 9.2), are not formalised

in this development; the general-field results are superseded by [20], which is formalised over every field.

Formalisation

The accompanying Lean 4 sources give a formal proof of Theorem A. The finite-set statement is `Pilz.pilz_conjecture` in `Pilz/Assembly.lean`. Build and verification commands are given in the repository's `README.md`.

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